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Environmental Exposure Assessment of Sulfonated Naphthalene Formaldehyde Condensates and Sulfonated Naphthalenes Applied as Concrete Superplasticizers

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Table of Contents

Summary

Sulfonated naphthalenes and their formaldehyde condensates (SNFC) are ubiquitous water pollutants due to their widespread uses, anionic character, and refractory behavior. They are predominantly applied in the textile industry and as concrete superplasticizers in the construction business, each with global uses of about 150'000 tons per year. The environmental impact of sulfonated naphthalenes originating from textile finishing was already thoroughly investigated. However, the role of the construction industry as a source for SNFC in the environment was largely unknown. This study focuses on the environmental risk of SNFC deriving from construction activities. Since modern construction technology is increasingly based on chemical admixtures and since SNFC are among the mostly used concrete admixtures this study is a suitable example for the environmental risk assessment of polar organic construction chemicals in general.

An **analytical method** for the determination of SNFC in environmental samples was developed. The method is based on synchronous excitation fluorimetry with a $\Delta\lambda$ of 105 nm. All SNFC components (monomers and oligomers with individual chainlengths) are recorded as a collective parameter. The samples were neither enriched nor the individual analytes separated. Interferences by humic acids and nitrate occurred only at concentrations higher than 1 mg C/L and 10 mg NO_3^-/L, respectively. Potential contribution of naphthalene to the fluorescence signal was prevented by extraction of the environmental sample with *n*-hexane. The detection limit was 0.2 µg/L.

Field studies were conducted on two tunnel construction sites. At both construction sites, cement suspensions containing SNFC were injected into the groundwater in order to stabilize the gravel of the aquifer in the surroundings of the tunnel. The groundwater was sampled for one and two years, respectively. SNFC concentrations of up to 230 µg/L were measured at a distance of about 5-10 m from the construction site.

The concentration of some SNFC components in the groundwater decreased faster than could be expected from sole dispersion of the SNFC plume. This enhanced decrease indicated biological transformation. **Biodegradation experiments** in the laboratory confirmed the observed

degradation behavior of the SNFC components in the field. Most monomers were degraded within 195 days, while naphthalene-1,5-disulfonate and the SNFC oligomers were refractory.

In the groundwater, SNFC components with chainlengths of one to four units were found. They exhibit different migration times in the aquifer. This was explained by **adsorption experiments** with a 0.16 to 0.5 mm fraction of aquifer material which showed that the adsorption ability of the oligomers was increasing with the chainlength. Oligomers with chainlengths of more than four units were found to adsorb within hours to an extent of more than 90 %. This implies that they are strongly sorbing onto the cement and, therefore, do not leach into the groundwater.

Investigations at the groundwater of the two field sites allowed to calculate the SNFC **mass fluxes** of the construction sites. It was estimated that about 5 % of the applied SNFC leaches into the groundwater. About 80 % of this amount is biodegraded in the aquifer, while 20 % remain in the groundwater.

Further investigations on **additional SNFC sources** revealed that SNFC from construction site wastewater as well as from wastewater of SNFC producers can enter the environment by way of wastewater treatment plants. Since naphthalene-1,5-disulfonate and the SNFC oligomers persisted in adapted biological treatment plants, they can enter the environment unchanged.

The measured environmental concentrations of SNFC were the basis for a **risk assessment** for groundwater and surface waters. Only one PEC/PNEC ratio for a worst case scenario turned out to be above 1, implying possible impact of the environment.

With this study, the major sources for SNFC in the environment which are related to the construction industry were identified and the fate and behavior of SNFC in the groundwater was thoroughly investigated, providing a representative example for the environmental risk of polar organic construction chemicals.

Zusammenfassung

Naphthalinsulfonate und Naphthalinsulfonatformaldehydkondensate (NSFK) sind Zivilisationschemikalien, welche aufgrund ihres ionischen Charakters, ihrer Persistenz und ihrer weitverbreiteten Anwendung die Gewässer verschmutzen. Sie werden vorwiegend in der Textilindustrie und als Hochleistungsbetonverflüssiger im Bauwesen angewendet. Für beide Anwendungen wird der weltweite Jahresverbrauch auf je 150'000 Tonnen geschätzt. Das Umweltgefährdungspotential sulfonierter Naphthaline, die in der Textilindustrie zum Einsatz kommen, wurde bereits eingehend untersucht. Die Rolle der Bauindustrie als Quelle für NSFK in der Umwelt ist jedoch noch weitgehend unbekannt. Der Schwerpunkt dieser Studie besteht darin, das Umweltrisiko von NSFK aus der Bauindustrie zu ermitteln. Da die moderne Bautechnologie vermehrt auf chemische Zusatzmittel angewiesen ist und NSFK zu den meist angewendeten Betonzusatzmitteln gehören, representiert diese Arbeit ein Musterbeispiel für die Abschätzung des Umweltrisikos von polaren organischen Bauchemikalien im Allgemeinen.

Eine **analytische Methode** zur Bestimmung von NSFK in Wasserproben wurde entwickelt. Diese basiert auf synchroner Anregungsfluorimetrie mit einem $\Delta\lambda$ von 105 nm. Alle NSFK-Komponenten (Monomere und Oligomere mit verschiedenen Kettenlängen) werden als Summenparameter erfasst. Die Proben wurden dazu weder angereichert noch aufgetrennt. Interferenzen durch Huminsäuren und Nitrat wurden nur bei Konzentrationen über 1 mg C/L und 1 mg NO_3^-/L beobachtet. Eine mögliche Verstärkung des Fluoreszenzsignals durch Naphthalin wurde durch das Extrahieren der Umweltprobe mit *n*-Hexan verhindert. Die Nachweisgrenze der Methode beträgt 0.2 µg/L.

An zwei Tunnelbaustellen wurden **Feldstudien** durchgeführt. Bei beiden Baustellen wurden NSFK enthaltende Zementsuspensionen in den Grundwasserleiter injiziert, damit der Schotter rundum den Tunnelquerschnitt stabilisiert werden konnte. Das Grundwasser wurde während eines, bzw. zweier Jahre überwacht. NSFK-Konzentrationen von bis zu 230 µg/L konnten in einem Abstand von 5-10 m zur Baustelle gemessen werden.

Die Konzentrationen von einigen NSFK-Komponenten in den Grundwasserproben sanken schneller als aufgrund der Verdünnung des

NSFK erwartet werden kann. Diese beschleunigte Abnahme wurde der biologischen Transformation zugeschrieben. Dies konnte durch **Bioabbau-experimente** im Labor bestätigt werden. Die Experimente zeigten, dass die meisten NSFK-Monomere innert 195 Tagen abgebaut werden. Hingegen sind Naphthalin-1,5-disulfonat und die NSFK-Oligomere persistent.
Im Grundwasser konnten NSFK-Komponenten mit Kettenlängen von eins bis vier Einheiten gefunden werden. Ihre Migrationszeiten im Aquifer waren unterschiedlich. Dieses Verhalten konnte mit **Adsorptions-experimenten** erklärt werden. Die Experimente zeigten, dass die Adsorptionsfähigkeit der Oligomere mit zunehmender Kettenlänge ansteigt. Oligomere mit Kettenlängen von mehr als vier Einheiten adsorbieren innert wenigen Stunden zu mehr als 90 %. Daraus kann gefolgert werden, dass sie stark an den Zement adsorbieren und daher nicht ins Grundwasser ausgewaschen werden.
Basierend auf den Grundwasseruntersuchungen der zwei Feldstudien war es möglich, die **Massenflüsse** dieser Baustellen zu berechnen. Die Abschätzungen zeigten, dass ca. 5 % des eingesetzten NSFK ins Grundwasser ausgewaschen wurde. Etwa 80 % davon können im Grundwasserleiter biologisch abgebaut werden, 20 % jedoch bleiben im Grundwasser. Untersuchungen an **weiteren NSFK-Quellen** zeigten, dass auch NSFK von Baustellenabwässern und Abwässer von NSFK Produzenten über Kläranlagen in die Umwelt gelangen können. Da Naphthalin-1,5-disulfonat und die NSFK-Oligomere auch in adaptierten biologischen Klärstufen nicht abgebaut werden, können sie die Kläranlage unverändert passieren.
Die gemessenen Umweltkonzentrationen von NSFK dienten dazu, eine **Risikoabschätzung** für NSFK im Grundwasser und in Oberflächengewässern zu machen. Ein PEC/PNEC Verhältnis basierend auf einem Worst-Case-Szenario war grösser als 1. Ein möglicher Umwelt-einfluss durch NSFK kann daher nicht ausgeschlossen werden.
Mit dieser Studie konnten die bauindustriespezifischen Quellen für NSFK in der Umwelt bestimmt und das Schicksal und Verhalten von NSFK im Grundwasser erforscht werden. Die Studie ist ein Beispiel für die Abschätzung des Umweltgefährdungspotential von polaren organischen Bau-chemikalien.

1

Introduction

It is generally agreed that clean water is a very precious good, probably the most important and fundamental natural resource. Moreover, the global water demand keeps rising. Not only in arid countries but also in Europe the water resources have to be protected from all kinds of negative impacts to guarantee enough water of good quality for the following generations. But with increasing industrialization and the subsequent use of chemicals, xenobiotic substances have been appearing in the environment. Some of these anthropogenic chemicals are well water-soluble and therefore often end up in the aquatic environment. There, they have a huge potential for negatively affecting the quality of water. Only around 1960, when detergents caused foam in natural waters, people became aware of the adverse effects of anthropogenic chemicals on the aquatic environment. Hence, the environmental fate and behavior of water-soluble chemicals became more and more the focus of scientific investigations. The occurrence of these chemicals in natural waters was monitored in order to identify their sources, assess their environmental risk and to propose measures to prevent negative impacts on aquatic life and human health.

Aromatic sulfonates are among the hydrophilic chemicals which were the subject of many environmental studies. The investigations were motivated by the fact that these compounds have many different applications in consumer products (e.g. as anionic surfactants in laundry detergents) and for industrial purpose (e.g. as dispersants and wetting agents in the textile industry). Moreover, their high production volumes led to the conclusion that they might appear in the aquatic environment. Their pathways into the environment and their behavior in sewage treatment plants and the environment have been the focus of research for quite some time at EAWAG [1-4]. In particular, sulfonated naphthalenes were found in rivers and contaminated groundwater (see Table 1.1), leading to questions regarding their sources. Besides their application for textile finishing, sulfonated naphthalenes are predominantly used as superplasticizers in the construction industry, with equal amounts used for the two applications. Each industry's global use amounts to up to 150'000 t/a [5]. Textile production is known to be a source for sulfonated naphthalenes in surface water, since some of them were found to be persistent in sewage treatment plants [2] and are thus released into the ambient surface waters. However, the role of the construction industry as a source for sulfonated naphthalenes in the environment has not been investigated yet. EAWAG set up a project on the release of sulfonated naphthalenes from construction activities into the environment. The umbrella organization of the cement producers participated since the general sensibilization for environmental issues asked for knowledge about the environmental impact of cement and concrete. This includes construction chemicals such as sulfonated naphthalenes added in increasing amounts to the cement and concrete. Hence, the cement industry felt obliged to answer questions regarding the relevance of construction activities as sources for sulfonated naphthalenes in the environment and to study the leaching, fate and behavior of sulfonated naphthalenes as a representative for water soluble construction chemicals. The aim of the presented work is to investigate these topics.

Table 1.1: *Environmental Concentrations of Sulfonated Naphthalenes [3, 6, 7]*

	Concentration [μg/L]		
Analyte[a)]	River Rhine	River Bormida	Contaminated groundwater
N15S	0.72		2
N27S	0.09	3460	5.7
N16S	0.25		11.9
N2S	0.14	320	34

[a)] N15S = naphthalene-1,5-disulfonate, N27S = naphthalene-2,7-disulfonate, N16S = naphthalene-1,6-disulfonate, N2S = naphthalene-2-sulfonate.

1.1 Construction Chemicals

1.1.1 Overview

The construction sector is a major industry in Switzerland, where investments in 1999 accounted for 8.3% of the national GDP (gross domestic product) [8]. The resulting consumption of concrete is 40 million tons per year (1991), implying a high consumption of concrete admixtures as well. In 1992 the turnover of concrete admixtures of Swiss Contractors in Switzerland was 15'000 t, qualifying them as high production volume chemicals [9].

The consumption of concrete admixtures is connected to modern construction technology, which requires concrete with very specific properties. These requirements can be fulfilled by adding chemicals. Concrete admixtures influence properties like consistency, pumpability, setting behavior, compressive strength and flowability. Today a standard concrete consists of 82 % aggregates, 12 % cement, 6 % water and *0.1 % concrete admixtures*. The large variety of concrete admixtures is divided into four main groups [9, 10]:

- **Set Retarding Admixtures:** Their use is often connected to ready mixed concrete which is transported to the site of application. They retard the hydratation of the cement (setting) during the transport time due to influence on the chemical reaction of the cement with water. Set retarding admixtures are based on saccharose, hydroxycarbonic acids, gluconates, phosphates and phosphonates.
- **Set Accelerating Admixtures:** They are added to shotcrete (concrete which is sprayed onto surfaces) for the fixation of slopes and freshly burst tunnel ceilings. For these applications the concrete must harden immediately. Set accelerating admixtures influence the chemical reaction of cement with water. They consist of calcium formiate, calcium nitrate, aluminium sulfate, alcali carbonates, -hydroxides and -aluminates.
- **Water Resisting Agents:** They are merely used for water reservoirs and structures in the groundwater. They decrease the permeability of the concrete by narrowing or closing pores or increasing the hydrophobicity of the concrete. For the application as water resisting agents lignosulfonates are combined with proteins, clay or bentonite.
- **(Super)plasticizers:** They are crucial admixtures for basic concrete technology, because many applications require a pumpable concrete. The flowability could be increased by adding water. However, because the excess water can not be chemically bound during hydratation of the cement, this would lead to water encapsulations and would lower the compressive strength of the concrete due to increased porosity. The addition of 1 % (w/w) of superplasticizers (relative to the amount of cement) has the same effect as the addition of 20 % of water, but

without producing pores. The plasticizing effect of superplasticizers is based on the dispersion of cement grains (Figure 1.1). The negatively charged superplasticizer molecules are adsorbed on the surface of the cement grains. The electrostatic repulsion between the negatively charged cement grain surfaces lowers the internal friction in the concrete and enhances the flowability of the concrete. Superplasticizers are based on sulfonated naphthalene-formaldehyde condensates, sulfonated melamine-formaldehyde condensates and lignosulfonates. Newer generations of superplasticizers consist of polycarboxylates.

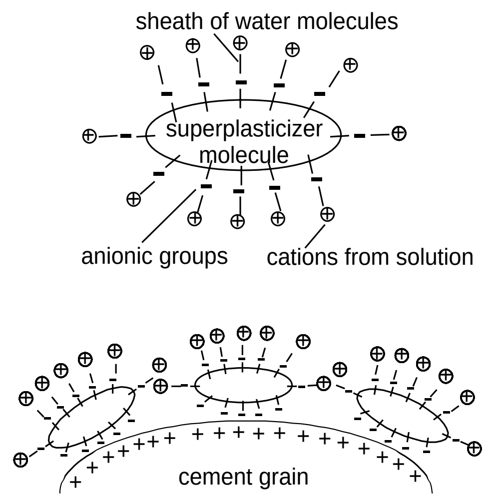

Figure 1.1: *Mechanism responsible for the superplasticizing effect of superplasticizers (from [11]).*

Superplasticizers were the most used concrete admixtures. In 1992, they amounted to 30 % of the totally consumed concrete admixtures in Switzerland. Of this amount about one third were superplasticizers based on sulfonated naphthalene formaldehyde condensates (SNFC), adding up to 10 % of the total use of concrete admixtures in Switzerland [9]. Recently, SNFC were used at three of the biggest tunnel construction sites in Switzerland, namely the Gotthard base tunnel and the Zurich-Thalwil

tunnel for the Swiss Federal Railways and the third Baregg tunnel for the highway Zurich-Bern.
The dominant role of SNFC in construction technology and the consumption figures qualify SNFC as an important representative of construction chemicals.

1.1.2 Construction Chemicals as Potential Sources of Environmental Contamination

Construction chemicals are often polar organic compounds (see above). Considering their polarity, they are assumed to leach from concrete into the aquatic environment. Since concrete is a prevalent construction material, the possible sources for environmental contamination with construction chemicals are widely spread. Construction chemicals could enter the environment through manufacturing plants, construction sites and through leaching from existing structures like buildings, roads, and demolition debris (see [12]).
In first attempts to assess the environmental risk of construction chemicals, laboratory leaching tests were made. In these tests the potential of concrete admixtures and other potentially harmful constituents of concrete, such as heavy metals and salts, to leach into water were investigated. The laboratory tests showed that the amount of e.g. superplasticizers leaching from hardened concrete was low and the leaching was restricted to the surface layer of the concrete [5, 13, 14]. These tests are reliable tools to investigate the compatibility of concrete with drinking water distribution systems (e.g. reservoirs) and to assess the hazard potential of demolition debris containing construction chemicals. It could also be shown that hardened concrete can most probably not be the source of construction chemicals in the environment.
The situation is different when fresh cement is in contact with water. Structures in ambient waters are often found at tunnel construction sites and for underground work in areas with high groundwater levels. The risk potential of such cases has not systematically been evaluated although severe environmental problems occurred in 1997 at two tunnel construction sites in northern Europe where construction chemicals

leached into ambient waters. In both cases, seeping of groundwater into the tunnel was stopped with injections of grouting agents in combination with cement. Symptoms characteristic for exposure to the toxic grouting agent based on acrylamide were observed in tunnel workers. In one of the two cases the leached grouting agent exfiltrated into the surface water. Poisoning of fish and cattle was observed and several water wells were contaminated [15-17].

Possibly, also production wastewater and wastewater from construction sites can be relevant sources for construction chemicals in the environment. Since in these cases the chemicals are not immobilized in the cement matrix, relevant amounts could reach the environment. In fact, earlier investigations showed the release of construction chemicals into the environment through wastewater [7, 12, 18].

It can be inferred that for an overall environmental risk assessment of construction chemicals the sources "application" and "production" have to be investigated thoroughly. These sources are expected to lead to the highest environmental concentrations and, depending on the toxicity of the chemicals, the environment could be impacted. Hence, this study focuses on construction sites in groundwater, construction site wastewater and on the wastewater from manufacturing plants.

1.1.3 Characterization of Sulfonated Naphthalene Formaldehyde Condensates (SNFC)

In this study, SNFC were chosen as representatives for water-soluble construction chemicals. Their characterization is important for estimating their potential environmental risk. SNFC are synthesized by unspecific condensation of sulfonated naphthalenes with formaldehyde. The resulting technical product contains oligomers as well as uncondensed monosulfonated and disulfonated monomers (see Figure 1.2). Technical SNFC mixtures are applied as dispersants, superplasticizers and tanning agents. The SNFC products used as superplasticizer are aqueous solutions containing about 15 % SNFC. The latter consists of 10 % monosulfonated monomers, 25 % disulfonated monomers and 65 % oligomers with chainlengths of up to 15 units. This composition was found in two investigated superplasticizer products and also in a product used by the textile industry.

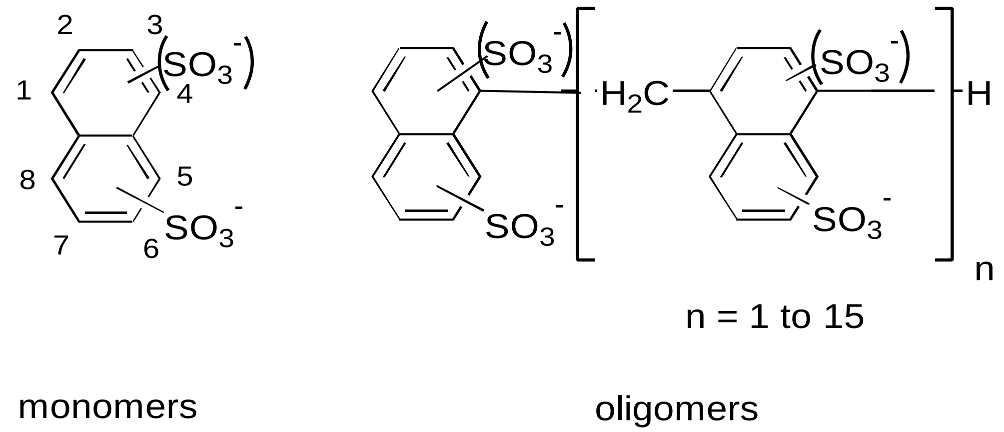

Figure 1.2: *Components of the technical SNFC mixture.*

The environmentally relevant properties of SNFC are not well known, but data for sulfonated naphthalenes exist. They have very low pK_a-values (<1) and, therefore, are fully deprotonated in ambient waters. Their ionic character leads to good water-solubility and low octanol water coefficients (log $K_{ow} \approx -1$) [19, 20]. They hardly tend to accumulate in organic matter or in living organisms. However, they are expected to be highly mobile in aquatic systems. The higher condensed oligomers form micelles when in aqueous solutions (see Figure 1.1). The critical micelle concentration (threshold concentration for the formation of micelles) for SNFC is around 0.4 mM [12], which is far above determined environmental concentrations. Generally, aromatic sulfonates have a low toxicity. They are rapidly excreted due to their hydrophilic character. In material safety sheets of technical SNFC mixtures LD_{50} values for rats were reported to be higher than 5'000 mg/kg body weight (based on solids) and LC_{50} values for fish (rainbow trout) were between 100 and 500 mg/L [21, 22]. Most aromatic sulfonates show no mutagenic or cancerogenic effects [20]. The biodegradability of sulfonated naphthalenes was investigated in the laboratory, in wastewater treatment plants and in the groundwater. It was generally found that monosulfonated naphthalenes are readily

biodegradable, while disulfonated naphthalenes are slowly degraded or persistent. SNFC oligomers are persistent as well. The degradable sulfonated naphthalenes are fully metabolized. No persistent transformation products are formed [2, 23-25].
In water works using bank filtrate water from the river Rhine, SNFC were found. While the dimeric SNFC were eliminated in the water purification plant after filtration with activated carbon, the monomeric components passed the activated carbon filter of a water work. Thus, existing water purification works do not in all cases eliminate aromatic sulfonates from drinking water [26].
Stability tests showed that chemical degradation of SNFC during the hardening of the cement (i.e. high pH and temperature) does not occur [12]. Hence, SNFC can be found unchanged in bleeding water and may be washed out of cement surfaces.
Although only a low ecotoxicological risk is attributed to sulfonated naphthalenes and their formaldehyde condensates, they are relevant water pollutants due to their mobility and persistence. Their amphiphilic character could possibly be hazardous for the environment, since their lipophilic part could associate with nonpolar contaminants and remobilize them into the aquatic environment.

1.2 Scope of this Work

The scope of this study was to investigate the main sources for SNFC in the aquatic environment and to study the behavior and fate of SNFC in ambient waters, especially groundwater.
Within this work the following studies were performed:

1) **Analytical Method (Chapter 2):** A compound specific analytical method is essential for investigations on contaminants in the environment. For the determination of SNFC a reliable method based on HPLC-FLD was published [27] and applied within this work. Since this method is time consuming and can only be applied by well-equipped laboratories, a rapid procedure for routine analysis which could be established in laboratories with limited funding was developed. This new analytical method is based on conventional

fluorimetry. Synchronous excitation mode allowed the determination of SNFC in groundwater samples without additional clean-up or analyte enrichment. The only preparatory step was the extraction of the environmental sample with *n*-hexane to prevent naphthalene from contribution to the fluorescence signal. Interferences occurred only at concentrations of humic acids and nitrate which are above the average concentrations found in the groundwater. The limit of detection was low enough for the determination of SNFC in groundwater samples.

2) **Field Studies (Chapters 3, 4 and 5):** In order to understand the sources of SNFC in the environment, field studies were conducted. The field studies focused on the source "application". The leaching behavior of SNFC out of fresh cement in contact with groundwater was investigated at two tunnel construction sites. The groundwater was monitored during one and two years, respectively. These studies revealed also the behavior of SNFC in the groundwater. Biodegradation and adsorption onto the aquifer material was observed. At a third tunnel construction site, grab samples from the construction wastewater and from the river receiving the wastewater were analyzed giving information on the contamination potential of construction wastewater.
3) **Biodegradation Experiments** (**Chapters 3 and 5**)**:** Laboratory degradation experiments aimed at explaining the observed biodegradation of SNFC in the aquifers at the two field sites. Therefore, sludge from the aquifer was incubated with SNFC and the obtained data was compared with field data. For the investigation of the biodegradability of SNFC in adapted systems, the industrial treatment plant of a SNFC manufacturing plant was sampled and the elimination of SNFC observed. With the sampling of the effluent of the industrial treatment plant, information on the SNFC source "production" were obtained.
4) **Adsorption Experiments (Chapter 4):** Since adsorption of SNFC was observed in the groundwater, the adsorption behavior of SNFC components was evaluated in laboratory experiments. Aquifer material was used as sorbent. The laboratory experiments could explain the leaching characteristics of SNFC out of fresh cement and information on adsorption processes onto aquifer material was obtained. The results

of the laboratory experiments were compared with the field observations.

5) **Mass Flux Calculations (Chapter 4):** The amount of leached construction chemicals at tunnel construction sites in the aquifer is of wide interest. Mass fluxes for SNFC were calculated for the two tunnel construction sites in the aquifer and compared.
6) **Risk Assessment (Chapter 5):** Within this study, environmental concentrations of SNFC were measured, providing exposure concentrations for aquatic organisms and humans up-taking drinking water contaminated with SNFC. In an environmental risk assessment these exposure data were related to effect concentrations of SNFC in order to evaluate the risk potential of the SNFC concentrations found in the field studies.

The measured environmental SNFC concentrations, the fate studies of SNFC in aquifers, and the evaluation of the main SNFC sources allowed an overall assessment of the environmental impact of SNFC, leading to practical implications for avoiding SNFC release into the environment.

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2

Analytical Methods for SNFC

Sulfonated naphthalenes and their formaldehyde condensates (SNFC) were determined in aqueous environmental samples by spectrofluorimetry. A clean-up step using n-hexane to extract possibly interfering nonpolar compounds such as naphthalene is the only preparatory procedure. Synchronous excitation mode with a $\Delta\lambda$ of 105 nm allows the determination of SNFC in environmental samples without additional clean-up or analyte enrichment. Interferences by humic acids and nitrate occurred only at concentrations higher than 1 mg C/L and 10 mg NO_3^-/L, respectively. The limit of detection was 0.2 µg/L, the average recovery was 104 % and the confidence interval (95 % certainty) was 24 %. The response factor for the quantitative determination of total SNFC, depending on the distribution of the different SNFC-components, was validated for groundwater from two field sites using an HPLC-FLD method as a reference method.

Ruckstuhl, S., Suter M. J.-F., Giger, W. (2001). *Rapid Determination of Sulfonated Naphthalenes and Their Formaldehyde Condensates in Aqueous Environmental Samples Using Synchronous Excitation Fluorimetry.* Analyst, 126 (11), 2072 - 2077.

2.1 Introduction

Sulfonated naphthalenes and their condensates with formaldehyde are widely used in industrial processes [1]. Quantitatively, sulfonated naphthalene products are most extensively used as tanning agents and as plasticizers for concrete, each with a global use of 150'000 t/a worldwide [2]. Concrete plasticizers are likely to become more important in the future, since modern concrete technology is based on an increasing use of synthetic organic admixtures.

Sulfonated naphthalenes are strong acids (pK_a-values < 1) and exhibit low octanol-water partition constants ($\log K_{ow} = -0.94$) [3, 4]. Thus, they are very highly soluble in water. Most of the sulfonated naphthalenes, e.g. disulfonated naphthalenes, are only slowly biodegradable [5, 6]. Due to these properties they are expected to be highly mobile and persistent in the aquatic environment. Although the toxicity of sulfonated naphthalenes is low (typically, fish toxicity LC_{50} = 100 - 500 mg/L) [7, 8], they may have some potential environmental impacts. Due to their amphiphilic character, sulfonated naphthalenes may act as remobilizing agents for nonpolar organic contaminants and in addition, unwanted disinfection-products could be formed during drinking water treatment.

The widespread and increasing use combined with the affinity to aquatic environments makes sulfonated naphthalenes a relevant pollutant of natural waters. Several authors reported on the occurrence of sulfonated naphthalenes in rivers and groundwater although their sources were unknown [9-11]. The reported concentrations are shown in Table 2.1.

Table 2.1: *Environmental Concentrations of Sulfonated Naphthalenes*

	concentration [µg/L]		
Analyte[a)]	River Rhine [9]	River Bormida [10]	Contaminated groundwater [28]
N15S	0.72		2
N27S	0.09	3460	5.7
N16S	0.25		11.9
N2S	0.14	320	34

[a)]Abbreviations see Table 2.3

However, there is no data on the occurrence of formaldehyde condensates in the aquatic environment near construction sites, where relatively high concentrations of the condensates, originating from concrete superplasticizers, are expected to be found. This topic gets more important, since the Swiss Authorities have issued a provisional SNFC threshold value of 1 mg/L for one construction site in a Swiss city where the groundwater serves as a reservoir for the water supply of the city.

Most presently available methods for the trace determination of sulfonated naphthalenes in aqueous samples focus on monomeric sulfonated naphthalenes. These methods are normally based on solid-phase extraction either with graphitized carbon black [11, 12] or with a C18 reversed-phase material using an ion pair reagent [9, 13, 14], followed by reversed-phase ion-pair liquid chromatography or capillary electrophoresis [15, 16] with UV or fluorescence detection. Some more recent publications report on methods based on mass spectrometric detection [17-19].

Wolf et al. published a method for the quantitative analysis of SNFC in aqueous environmental samples using an enrichment procedure with C18 material and reversed-phase ion-pair liquid chromatography with fluorescence detection [20]. A clean separation of the monomers and the oligomers is achieved. Total SNFC in the aqueous phase of fresh cement water was also determined using UV spectrophotometry and spectrofluorimetry, but the selectivity and sensitivity of these methods is not sufficient for environmental samples [21, 22].

Fluorescence spectroscopy is a powerful technique for the determination of low concentrations of fluorescent compounds such as SNFC. Because of its high sensitivity and selectivity, no enrichment of SNFC in non-fluorescent matrices is needed. The selectivity is based on the fact that relatively few compounds show intrinsic fluorescence and that the emission intensity depends on two variables, namely the excitation and emission wavelength. However, with conventional fluorescence spectroscopy the emission bands are very broad. Narrower bands can be obtained by scanning the excitation and emission wavelength synchronously [23]. Thus, with the synchronous excitation mode, the selectivity can be increased to such an extent that aqueous environmental and biological samples can be analyzed without prior separation or enrichment [24].

In this article we present an analytical method based on fluorescence spectroscopy for the quantitative determination of the total amount of SNFC (monomeric sulfonated naphthalenes and condensates with formaldehyde) in aqueous samples. This method was developed in order to obtain a readily usable method, which yields reliable quantitative results within a short analysis time. Furthermore, this method can easily be established in laboratories with limited funding, thereby offering an alternative to analytical methods which are becoming more and more sophisticated and expensive. The method was validated using environmental samples. Low detection limits, good accuracy and precision as well as simple handling and instrumentation make this a rapid method ideally suited for routine analyses and hence monitoring of processes affecting the groundwater.

2.2 Experimental

2.2.1 Chemicals and Materials

Naphthalene-1-/-2-sulfonate (N1S, N2S), naphthalene-1,5-/-1,6-/-2,6-/-2,7-disulfonate (N15S, N16S, N26S, N27S) and diphenylamine-4-sulfonate were obtained from Fluka AG (Buchs, Switzerland), Aldrich (Steinheim, Germany) and TCI (Tokyo, Japan). Naphthalene-1,7-disulfonate (N17S) and the condensed naphthalene-2-sulfonate dimer (8,8'-methylenebis-2-naphthalenesulfonate) were kindly donated by Carmen Wolf, TZW Karlsruhe, Germany. In this report the "technical SNFC mixture" refers to Galoryl LH 120 from CFPI Industries (Gennevilliers, France). Tetrabutylammonium bromide (TBABr, puriss.) and formaldehyde solution (37 % w/w) were obtained from Fluka AG. Sodium nitrate was from Merck (Darmstadt, Germany) and humic acid (sodium salt) was from EGA-Chemie. n-hexane (p.a.) was purchased from Merck. HPLC-grade water, acetonitrile (ACN) and methanol (MeOH) (both multisolvent®) were purchased from Scharlau Chemie S. A. (Barcelona, Spain). The C18-material for solid-phase extraction LiChrolute RP-18 (40 - 63 µm) was from Merck, the 6 mL solid-phase extraction tubes (polypropylene) and polyethylene frits were from Supelco SA (Bellefonte, USA). Cellulose nitrate and regenerated cellulose membrane filters with 0.2 µm pore size were purchased from Sartorius GmbH (Göttingen, Germany).

2.2.2 Sample Collection and Treatment

Two Swiss construction sites were investigated. At field site A (Zurich-Thalwil), a railroad tunnel of a total length of 9.4 km is being built. The first section of the tunnel is embedded in alluvial material consisting of gravel and sand deposition. Injections with a cement suspension were made over a length of about 400 m to stabilize the gravel around the tunnel. For these injections the cement was fluidized with SNFC which was partially washed out, due to the groundwater level being above the tunnel. Samples were taken from several piezometers downstream of the construction site at a distance of 25 to 130 m. Field site B (Baregg) is a tunnel construction site for a national highway. A stagnant body of groundwater lays above one part of the tunnel. Cement was injected into the gravel above the tunnel over a length of 150 m in order to stabilize the soil prior to construction. The cement was fluidized with SNFC, some of which partially leached into the groundwater. The groundwater was sampled through three piezometers situated at a distance of about 10 m from the future tunnel.
Groundwater samples from construction site A were collected in amber glass bottles and stored at 4°C. The samples from construction site B were also collected in amber glass bottles and stabilized on site by adding 1 % (v/v) formaldehyde solution.

2.2.3 Synchronous Excitation Fluorimetry

Highly concentrated samples were diluted using double distilled water. 10 mL of sample were extracted in a separatory funnel with an aliquot of *n*-hexane (10 mL) to remove nonpolar substances, especially naphthalene. The fluorescence signal of the latter interferes with the signal of the sulfonated naphthalenes. The water fraction was then put into a cuvette and degassed for 5 min with nitrogen to remove oxygen, which could otherwise quench the signal. The samples were measured immediately with a Perkin Elmer Model LS-3 fluorescence spectrometer (Perkin Elmer, Rotkreuz, Switzerland) with spectral bandwidths of 10 nm. The cuvettes had a light path length of 10 mm and were from Huber & Co. AG, Rheinach, Switzerland. The spectra were acquired and integrated using a program designed on LabVIEW 4.0 for Macintosh from National Instruments, Ennetbaden, Switzerland.

The samples were measured in synchronous scan mode with a $\Delta\lambda$ of 105 nm. The excitation wavelength was scanned from 200 - 400 nm, the emission wavelength accordingly from 305 - 505 nm. The attenuation was 0.3 and the scan speed 120 nm/min.

A naphthalene-2,6-disulfonate solution (4.82 µg/L) was measured at the beginning and the end of a sample series to correct for intensity changes of the xenon-lamp. The measured peak areas of the samples were divided by the average of the fluorescence intensity of the naphthalene-2,6-disulfonate solution before and after measuring the sample series.

For the estimation of the limit of detection a sample with an SNFC concentration below 1 µg/L was extracted and measured ten times. The limit of detection was calculated as three times the standard deviation of the ten measurements.

The recovery was calculated analyzing a spiked groundwater sample (technical SNFC mixture: 1.75 µg/L) by four replicate determinations.

The confidence interval was obtained by using the standard deviation of the measurement of a real groundwater sample from field site A (2.1 µg/L, eight replicates). The standard deviation was multiplied by the student-t-factor for 95 % certainty.

For the experiments with humic acid a stock solution of 1 mg/ml humic acid was prepared. Its DOC concentration was 0.5 mg/ml. The DOC-specific absorption coefficient at the wavelength 280 nm (a*) was 4.8 L/(mg m).

For the experiments with nitrate a stock solution of 0.15 mg/ml nitrate was prepared.

2.2.4 HPLC/UV/FLD

Environmental samples were prepared and measured according to the procedure developed by Wolf et al. [20], which is a reliable and robust method. A cartridge with 1 g of solid-phase material (LiChrolute) was conditioned with 5 mL of MeOH and 10 mL of HPLC-grade water. Samples, with a total volume of 500 mL after the addition of 1 mmol/L TBABr and 2.5 µg/L of internal standard (diphenylamine-4-sulfonate), were percolated through the cartridge in about 1 h. The cartridges were dried with ambient air for 1 h under a vacuum and then eluted with 5 mL acetonitrile. The solvent was evaporated with nitrogen, then the dry residues were redissolved in 500 µL of a 4 mM TBABr aqueous solution and 50 µL were injected onto the HPLC column. Liquid chromatography was performed using an HP 1090L Series II liquid chromatograph (HPLC) equipped with a diode array detector (HP 1090) and a programmable fluorescence detector (FLD) (HP 1046) from Hewlett Packard AG (Switzerland). An octadecylsilica column (Hypersil ODS, 5 µm, 250 x 4 mm i.d.) with a precolumn (5 x 4 mm i.d.) was used (Macherey-Nagel AG, Switzerland). The aqueous mobile phase (A) was HPLC-grade water with 4 mM TBABr and was filtered through a 0.2 µm cellulose nitrate filter. A 25/75 mixture of water and acetonitrile with 4 mM TBABr was used as organic modifier (B). Both mobile phases were degassed during 20 - 30 minutes by a stream of helium at the start of a measurement series. The following gradient elution was used: 0 min 30 % B, 40 min 35 % B, 80 min 80 B and 85 min 100 % B isocratic to 95 min and back to the initial conditions at 100 min followed by 2 min of equilibration. The flow rate was 0.8 mL/min, column temperature was held at 40°C, and the injection volume was 50 µL. The UV-absorption wavelength was 300 nm and 220 nm with a reference wavelength of 550 nm. 300 nm is the absorption wavelength of the internal standard. The FLD excitation wavelength was 230 nm, the acquired emission wavelength was 360 nm, which is the preferable wavelength for the SNFC-oligomers. The photomultiplier gain was 9. The cut off filter absorbed the emission wavelengths up to 335 nm. The substances were identified based on their absorption and emission wavelengths and their retention times in the FLD chromatogram (see Table 2.2). The standard deviation of the retention times (Table 2.2) is due to the method with a very slow gradient. With the different matrices from the different field sites and with the increasing age of the chromatography column the retention times got a increasing standard deviation. To be sure about the identity of the analyte the samples were spiked with a standard mixture. The concentrations of the individual compounds were estimated by external calibration. The calibration of the dimer can also be used for the other oligomers due to the independence of the response factors (slope of the calibration curve) from the degree of polymerization [20]. The intensity of the FLD-lamp was monitored with an external FLD-standard (naphthalene-2,6-disulfonate). For validation of the method, recovery, precision, limit of detection and limit of quantification were determined and compared with the results obtained by Wolf et al. [20]. Each environmental sample was enriched and analyzed twice.

Table 2.2: *Retention Times and Their Standard Deviation (n=8) of the SNFC Compounds in Enriched Groundwater Samples Spiked with the Technical SNFC Mixture*

	retention time [min]
naphthalene-1-sulfonate	32.87 ±0.25
naphthalene-2-sulfonate	35.87 ±0.24
naphthalene-2,6-disulfonate	36.85 ±0.30
naphthalene-1,5-disulfonate	38.33 ±0.29
naphthalene-2,7-disulfonate	40.46 ±0.31
naphthalene-1,6-disulfonate	42.27 ±0.31
naphthalene-1,7-disulfonate	51.85 ±0.18
condensate n=2	63.16 ±0.12
condensate n=3	68.31 ±0.15
condensate n=4	71.15 ±0.09
condensate n=5	73.27 ±0.07
condensate n=6	74.89 ±0.08
condensate n=7	76.19 ±0.06

2.3 Results and Discussion

2.3.1 Method Validation

Acquisition with synchronous scan from 305 to 505 nm emission wavelength ($\Delta\lambda$ = 105 nm) gives a fluorescence signal for SNFC with a maximum emission intensity between 330 and 340 nm.

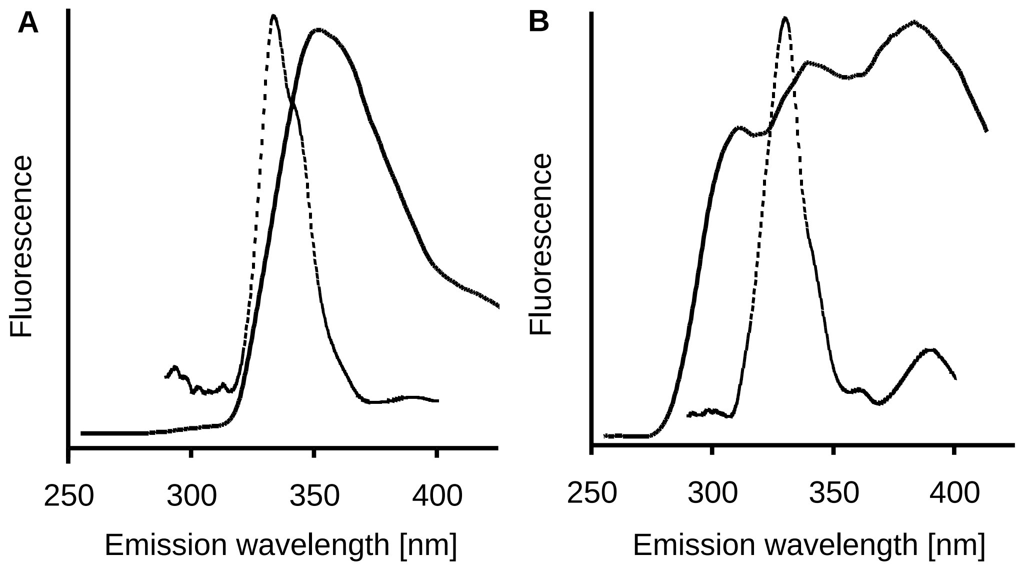

Figure 2.1: *A: Emission (λ_{ex} = 230 nm) and synchronous scan spectrum ($\Delta\lambda$ = 105 nm, broken line) of total SNFC (technical mixture) in aqueous solution. B: Environmental sample with SNFC contamination measured with conventional fluorimetry (λ_{ex} = 230 nm) and with synchronous scan mode ($\Delta\lambda$ = 105 nm, broken line). The signal of the conventional fluorimetry is interfered by matrix effects. With the synchronous scan signal no interferences occur. The difference between the synchronous scan signal of Figure A and B is due to different relative concentration of the SNFC components in the technical mixture and the environmental sample.*

Figure 2.1 shows the advantages of the synchronous scan acquisition mode. The signal is much narrower than the one of the conventional emission spectrum. The probability for interferences with matrix compounds is much smaller and interferences with Rayleigh scattering can be avoided.

One major advantage of the analytical procedure is that a clean-up with *n*-hexane is the only preparatory measure, thus reducing the number of steps in the analysis in which analytes may be lost compared to other methods. Recovery experiments yielded an average recovery of 104 %. The confidence interval for 95 % certainty is 24 % in the concentration range of the environmental samples from field site A. The limit of detection is 0.2 μg/L which is more than three orders of magnitude below the provisional threshold for groundwater issued by the Swiss authorities (1 mg/L).

2.3.2 Influence of Matrix

The robustness of the method towards possible matrix components such as humic acids and nitrate was thoroughly investigated. Humic acids can potentially enhance the fluorescence signal of a sample because they show some fluorescence. Using the synchronous scan method the fluorescence of humic acids does not interfere with the SNFC fluorescence [25]. However, humic acids show an inner filter effect. In order to quantify this effect a SNFC standard solution (30 μg/L technical SNFC mixture in double distilled water) was measured at different concentrations of humic acid. The threshold concentration, which had an impact on the fluorescence of SNFC was between 0.5 and 1 mg C/L (Figure 2.2). Furthermore, the clean-up had no influence on the SNFC-fluorescence, because humic acids are not well extracted by *n*-hexane. In this study, humic acids could be neglected because humic acid concentrations in Swiss groundwaters are generally below 1 mg/L. Furthermore, the DOC-specific absorption coefficient at wavelength 280 nm (a*) of dissolved organic matter in Swiss natural waters is 1.5 to 5 times lower than the one of pure humic acid [26].

Nitrate also shows an inner filter effect for the wavelength range used for SNFC measurements. The average concentration of nitrate in the groundwater of the field sites was around 15 mg/L. In order to check the influence of nitrate, a SNFC standard solution (30 µg/L technical SNFC mixture in double distilled water) was spiked with different nitrate concentrations. Only a slight effect at concentrations below 10 mg/L could be observed (Figure 2.2). However, at concentrations higher than 10 mg/L the fluorescence was increasingly filtered out.

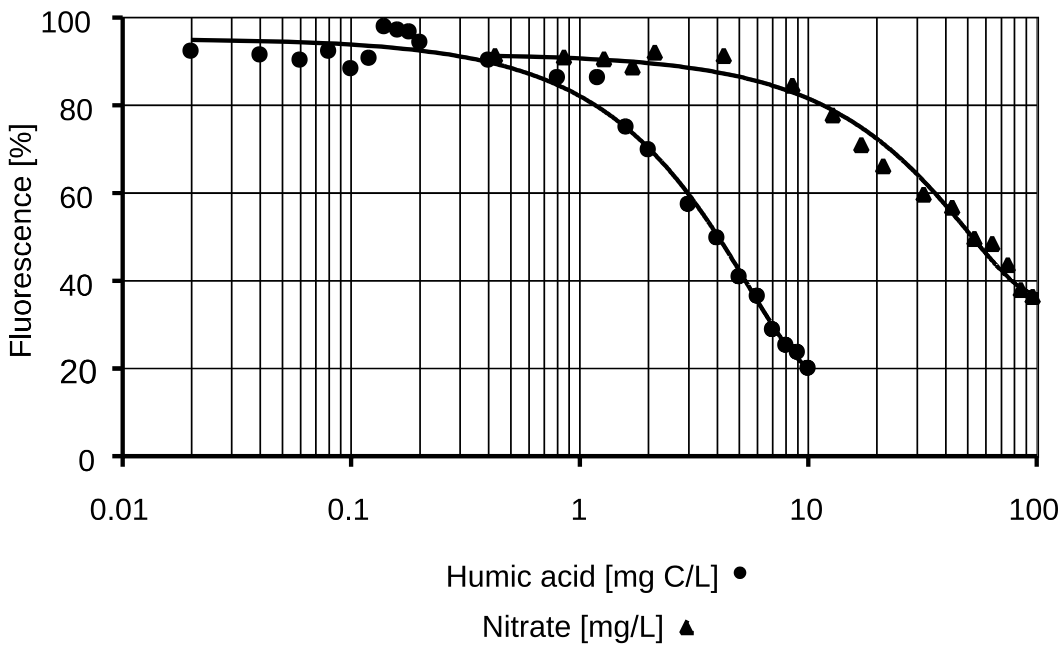

Figure 2.2: *Influence of humic acid concentration (expressed as organic carbon concentration) and nitrate concentration on the fluorescence of SNFC.*

When comparing the influence of humic acid and nitrate on the SNFC fluorescence, it can be observed that the slope of the decrease of the fluorescence intensity with humic acid is steeper. This is due to the broad absorption band of humic acids covering the whole spectral range which suppresses the SNFC fluorescence over the whole signal. The effect of nitrate is minor because the significant absorption band of nitrate is much narrower with the maximum at 220 nm and interferes only with the first part of the SNFC signal.

No pH-effect on the fluorescence of SNFC could be observed. This can easily be explained by the very low pK_a of the sulfonic acid groups (< 1), which excludes protonation. Furthermore, the pH of environmental samples is usually around 7, or can be adjusted if needed.

2.3.3 Optimization of the Quantitation

The crucial requirement for measuring the total SNFC content of a sample with fluorescence spectroscopy is that all SNFC components have the same fluorescence properties and thus can be summarized by one single fluorescence peak. With the developed synchronous excitation fluorimetry all monomeric SNFC compounds and the dimer, as the representative of all oligomers, showed a signal with a maximum fluorescence intensity between 330 and 340 nm. However, there were some differences in the response factors (i.e. slope of the calibration curves). Table 2.3 shows that N2S was the most fluorescent monomer and therefore has the highest response factor. The response factors of N2S and N15S, the least fluorescent monomer, differed by a factor of 2.8.

The variable response factors of the different SNFC components hampered an accurate quantification, because the ratio of the various SNFC components in environmental samples varies. It was observed that the different physico-chemical properties and biodegradability have an influence on the behavior of the different SNFC components in the aquatic environment (see below), leading to different relative concentrations in comparison to the technical mixture. Thus, using an average response factor or the response factor of the technical mixture is not an optimum solution.

Table 2.3: *Calibration Curves for the Synchronous Excitation Fluorescence Method Obtained from Linear Regression (the Response Factor is Defined as the Slope of the Calibration Curve)*

Compound	Range [µg/L]	Response factor [area units/(µg/L)]	Correlation coefficient
naphthalene-1-sulfonate (N1S)	1.5 - 7.4	4.31	0.998
naphthalene-2-sulfonate (N2S)	0.9 - 6.2	5.46	0.999
naphthalene-1,5-disulfonate (N15S)	2.0 - 10.0	1.95	1.000
naphthalene-1,6-disulfonate (N16S)	1.0 - 5.0	2.20	0.997
naphthalene-1,7-disulfonate (N17S)	0.7 - 2.1	3.32	0.995
naphthalene-2,6-disulfonate (N26S)	0.6 - 4.5	3.77	0.999
naphthalene-2,7-disulfonate (N27S)	0.9 - 4.5	3.09	0.993
technical SNFC mixture	1.1 - 5.8	3.09	0.996
dimer	0.6 - 1.9	4.50	0.996
field site A	1 - 35	2.14	0.989
field site B	1 - 50	4.22	0.983

First, attempts to quantify total SNFC were made with a "best-worst-case-scenario". By assuming that either the whole fluorescence signal comes from the compound with the highest response (N2S) or from the compound with the lowest response (N15S) (see Table 2.3) it is inferred that the true value lies between the two extremes. In order to verify this assumption, 37 groundwater samples from piezometers of field site A with an unknown ratio of the SNFC components were analyzed by both the synchronous excitation fluorimetry method and the precise HPLC-FLD method ([20], see Experimental Section). Figure 2.3 shows the ranges of the "best-worst-case-scenario" compared with the values from the HPLC-FLD measurements. In all cases the HPLC-FLD values are within the range obtained from the "best-worst-case-scenario", when considering the confidence interval of the HPLC-FLD method (± 26 %). The accuracy of the fluorescence spectroscopy measurements itself has to be considered as high due to the good correlation of the SNFC concentration range obtained by fluorescence spectroscopy and the exact SNFC concentrations obtained by HPLC-FLD, which leads to the conclusion that matrix effects (see above) and cross-interferences were negligible. Thus, false negative or false positive results as known from screening tests do not appear with this method when not measuring below the quantification limit.

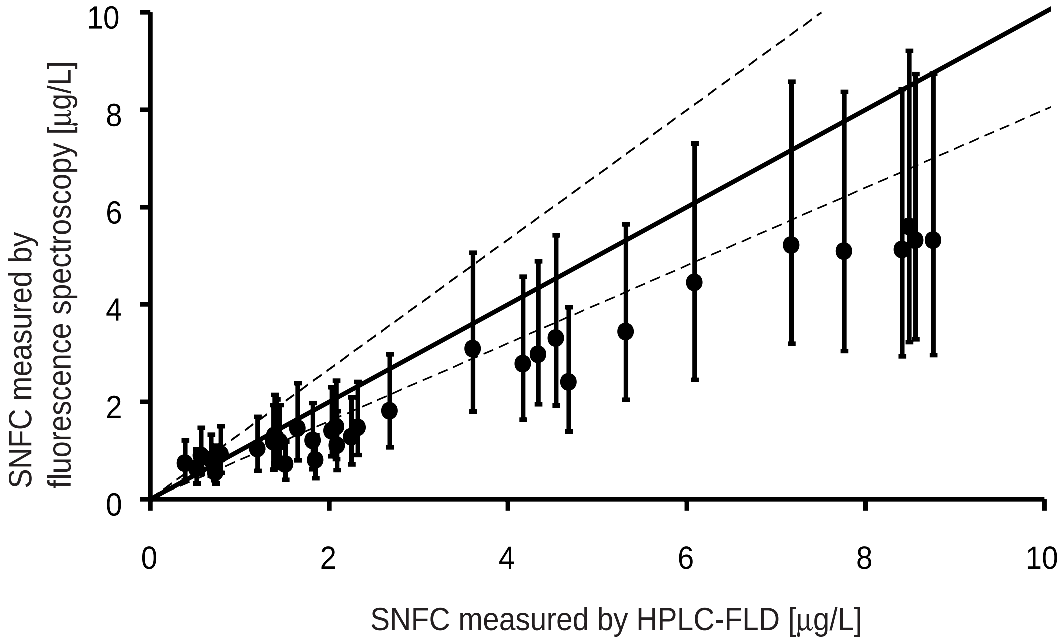

Figure 2.3: *Correlation of SNFC concentrations in groundwater samples from field site A measured by fluorescence spectroscopy and HPLC-FLD. The quantification with an average response factor is not accurate (•). With an accurate quantification method the values should have a slope of 1 (black line). The vertical error bars represent the ranges calculated by the "best-worst-case-scenario". Considering the confidence interval (± 26 %) of the HPLC-FLD method (---), the HPLC-FLD results are within the range of the "best-worst-case-scenario".*

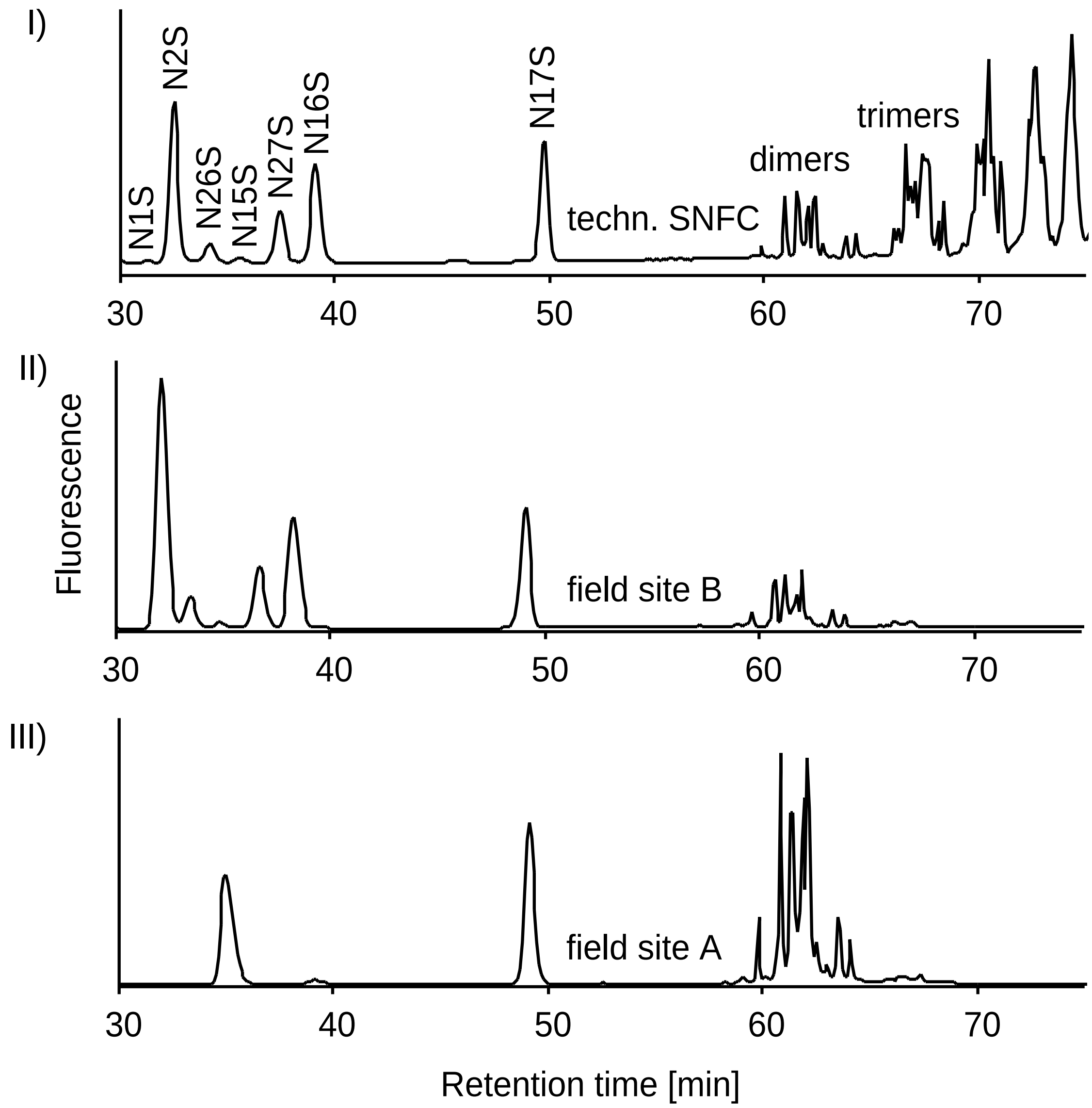

Figure 2.4: *Chromatograms reflecting the distribution of sulfonated naphthalenes and their formaldehyde condensates in different samples: I) technical SNFC-mixture, II) groundwater of field site B, III) groundwater of field site A. For abbreviations see Table 2.3.*

In order to obtain specific values instead of concentration ranges when using fluorescence spectroscopy, which would correlate well with the HPLC data, the processes causing changes in the composition of the SNFC compounds in groundwater must be evaluated. The HPLC-FLD chromatogram (Figure 2.4, Trace I) shows the distribution pattern of sulfonated naphthalenes and their formaldehyde condensates in the technical mixture. For the monomers the same pattern was found in the samples from field site B, where the samples were taken close to the injection site (Trace II). However, the distribution pattern of the monomers in the samples of field site A (Trace III) differed from the one in the technical mixture, i.e. only the monomers N15S and N17S are found. The groundwater from field site A was sampled 30 to 100 m downstream of the construction site. The groundwater needs a migration time between 1 to 3 months for this distance. Biodegradation can take place during this period and sorption is possible as well. This results in a typical distribution composed of only the monomers N15S and N17S, which can be explained by their persistence [6, 27]. Even after several months, they can still be found in the groundwater. The low response factor of N15S has a substantial impact on the overall response factor for the quantification of the samples from field site A. This is contrary to the groundwater from field site B, where the contribution of the N15S fluorescence to the overall response factor is only minor.

In the case of the condensates the analysis of field samples with the HPLC-FLD method showed that the more highly condensed oligomers are not leached from the cement. In the samples from field site B, oligomers up to n=4 were detected (not visible in Figure 2.4 because of the scale), while in the samples of field site A mostly dimeric condensates were found at a distance of 30 to 100 m. The distribution of the condensates in the samples of the two field sites had no major effect on the measured fluorescence because all condensates have a similar response factor.

The above-mentioned observations from the field studies have to be considered while developing a method for the SNFC quantification in environmental samples. It was decided to fit two response factors based on the samples from the two field sites (see Figure 2.5). Field site A gave a good fit with a response factor that can be expressed as 1.095 times the response factor of N15S. The optimum fit for field site B was obtained with 2.165 times the response factor of N15S. The curves have a correlation coefficient of 0.989 for A and 0.983 for B, respectively. With these fitted response factors good results were obtained during the whole sampling periods (12 and 6 months for field site A and B, respectively).

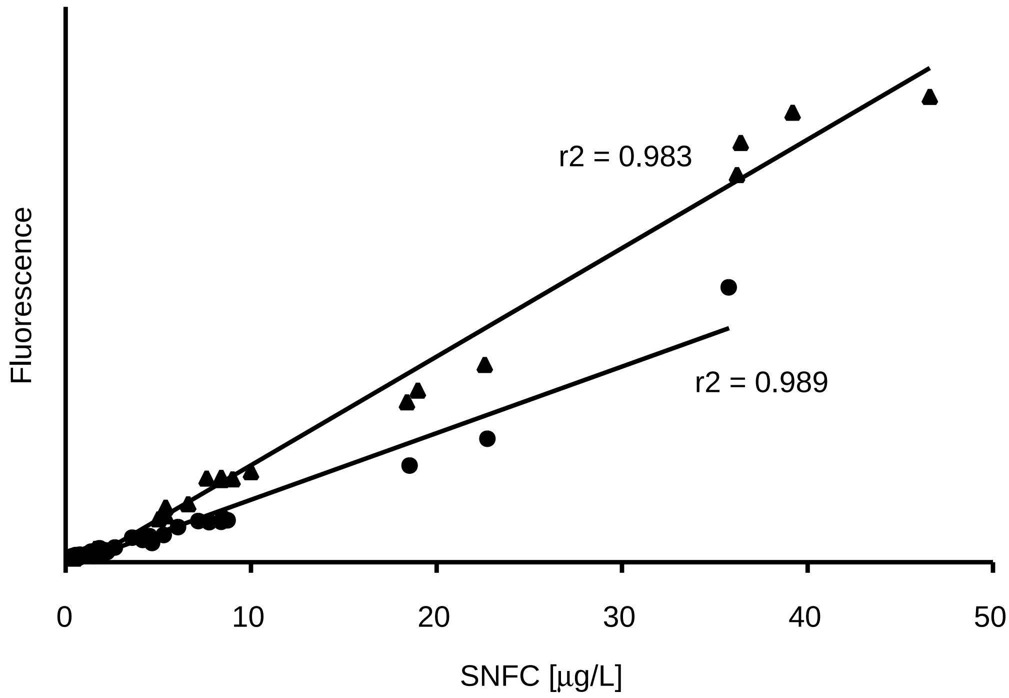

Figure 2.5: *The response factors for the field site A and B were obtained by plotting the fluorescence intensity from fluorescence spectroscopy as a function of the SNFC-concentration from the HPLC-FLD measurements. The response factor for field site A (•) and B were calculated from the slope of the trendlines.*

In order to apply the presented method to other field sites, the response factor needs to be estimated or determined by HPLC-FLD. The estimation can be made based on the migration time in the aquifer, with field site A and B representing the two extreme cases. With this additional information satisfactory results can be expected. Alternatively, a reliable confidence interval can be calculated using the "best-worst-case-scenario".

2.4 Conclusions

The presented method demonstrates, that organic compounds can be determined in complex matrices without sophisticated equipment and time-consuming sample preparation steps. The described method can be implemented in laboratories with only limited resources. Furthermore, results are obtained in less than half an hour. The detection limit reaches far below the provisional threshold value for SNFC in groundwater, which has been issued by the Swiss authorities. Thus, this method can easily be used to obtain quantitative results in exploratory screening studies.

Acknowledgments

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3

Leaching and Biodegradation of SNFC

Sulfonated naphthalene formaldehyde condensates (SNFC) and their monomeric analogues are used as concrete superplasticizers for tunnel construction through aquifers. The analyses of groundwater samples collected 5 m away from a construction site clearly indicated that SNFC leached into the groundwater. A maximum SNFC concentration of 230 μg/L was found and it was shown that only the monomeric sulfonated naphthalenes and the condensates up to the tetramer leached in substantial amounts. The decrease in concentration of several monomeric components could not be explained by mere dispersion, but rather indicates a biological transformation in the aquifer. This was confirmed at a second field site and by laboratory degradation experiments with piezometer sludge as inoculum. Lag and degradation phases for the sulfonated naphthalenes were 0 to 96 d and 3 to 143 d, respectively. Naphthalene-1,5-disulfonate and the oligomeric components were neither degraded in the aquifer nor in the laboratory experiments within an observation time of up to 195 d. This clearly indicates their persistence in subsurface waters.

Ruckstuhl, S., Suter M. J.-F., Kohler, H.-P. E., Giger, W. *Leaching and Biodegradation of Sulfonated Naphthalene Formaldehyde Condensates from Concrete Superplasticizers in Groundwater Affected by Tunnel Construction.* Submitted to Environ. Sci. Technol.

3.1 Introduction

Modern concrete construction technology uses increasing amounts of concrete admixtures, which are mostly synthetic organic compounds. Their global annual production volumes is on the order of several thousand tons. Many of these chemicals are very water soluble. Thus, a potential impact on the aquatic environment is inferred upon construction activities in or close to aquifers. Recently, two cases were reported in northern Europe, where construction chemicals leaching into the groundwater caused severe environmental problems [1-3].

Regarding production volumes, the most important representatives of water soluble concrete admixtures are superplasticizers based on sulfonated naphthalene formaldehyde condensates (SNFC). Their world-wide annual consumption is about 150'000 tons [4]. Technical SNFC mixtures consist of several mono- and disulfonated monomers and their condensed oligomers (Figure 3.1). Sulfonated naphthalenes and their formaldehyde condensates exhibit low octanol-water partition coefficients (log $K_{ow} = -0.94$) and low pK_a values (< 1) [5, 6]. Several publications report on the biodegradation of monomeric sulfonated naphthalenes [7-9]. Generally, the degradation pathway of sulfonated naphthalenes starts with desulfonation followed by ring cleavage. Disulfonated naphthalenes are less biodegradable than monosulfonated analogues. However, Nörtemann et al. [10] and Ruff et al. [11] reported on the degradation of disulfonated naphthalenes in the laboratory. They found that the second sulfonate group is eliminated after cleavage of the first ring. No information on the degradation and transformation of the condensed oligomers is currently available in the literature. The toxicity of SNFC is low (e.g. fish toxicity: $LC_{50} = 100 - 500$ mg/L) [12, 13]. Possible environmental effects of SNFC include the contamination of drinking water resources and the remobilization of nonpolar organic contaminants, due to their amphiphilic character.

SNFC are used for tunnel construction through aquifers, thus, SNFC are expected to be found in the groundwater near construction sites. However, there are no data published on the occurrence of SNFC under such circumstances. Swiss authorities have released a provisional standard for

SNFC in groundwater of 1 mg/L for one construction site in a Swiss city where groundwater serves as a resource for the water supply of the city.
This study focuses on the leachability of SNFC out of fresh cement which is in contact with groundwater. The fate of SNFC in the aquatic environment is evaluated with emphasis on biodegradation in aquifers. Therefore, groundwater samples from an aquifer affected by tunnel construction were analyzed and biodegradation was studied in laboratory experiments.

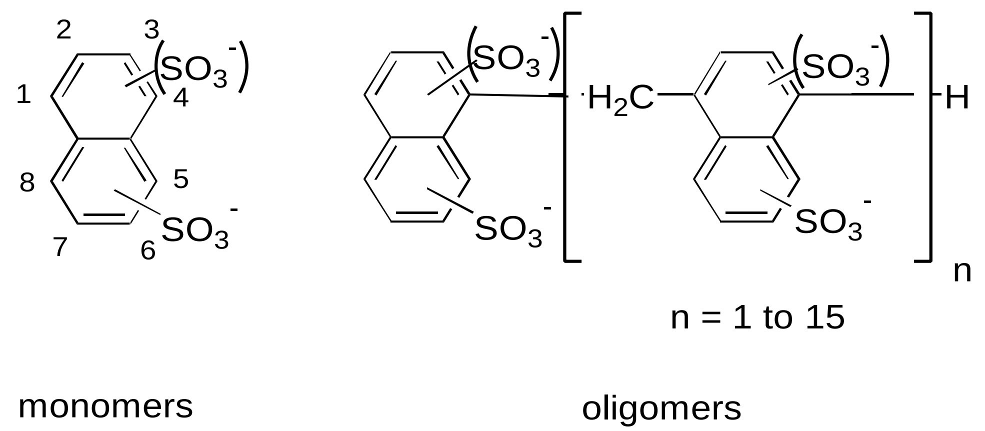

Figure 3.1: *Chemical structures of the components of technical SNFC mixtures: Monosulfonated and disulfonated naphthalenes (monomers) and formaldehyde condensates (oligomers).*

3.2 Experimental

3.2.1 Chemicals and Material

Naphthalene-1-/-2-sulfonate (N1S, N2S), naphthalene-1,5-/-1,6-/-2,6-/-2,7-disulfonate (N15S, N16S, N26S, N27S), and diphenylamine-4-sulfonate were obtained from Fluka AG (Buchs, Switzerland), Aldrich (Steinheim, Germany), and TCI (Tokyo, Japan). Naphthalene-1,7-disulfonate (N17S) and the condensed naphthalene-2-sulfonate dimer (8,8'-methylenebis-2-naphthalenesulfonate) were kindly made available by Carmen Wolf, TZW Karlsruhe, Germany. "Technical SNFC mixture" refers to Galoryl LH 120 from CFPI Industries (Gennevilliers, France). The technical SNFC mixture contains about 12 % SNFC, of which 11 % were monosulfonated monomers, 23 % were disulfonated monomers, and 66 % were formaldehyde condensates. Tetrabutylammonium bromide (TBABr, puriss.) and formaldehyde solution (37 % w/w) were obtained from Fluka AG. HPLC-grade water, acetonitrile (ACN), and methanol (MeOH) (both multisolvent®) were purchased from Scharlau Chemie S. A. (Barcelona, Spain). The C18-material for solid-phase extraction LiChrolute RP-18 (40 - 63 μm) was purchased from Merck, and the 6 mL solid-phase extraction tubes (polypropylene), and polyethylene frits were supplied by Supelco SA (Bellefonte, USA). Potassium hydrogen phosphate (KH_2PO_4), di-sodium hydrogen phosphate dihydrate ($Na_2HPO_4*2H_2O$), ammonium chloride (NH_4Cl), magnesium sulfate heptahydrate ($MgSO_4*7H_2O$), and calcium chloride dihydrate ($CaCl_2*2H_2O$) were all purum from Fluka AG (Buchs, Switzerland).

3.2.2 Sampling Sites

Field site A (Baregg) is a tunnel construction site for a Swiss national highway. A stagnant body of groundwater lays above part of the tunnel. Cement suspensions with SNFC were injected through drilled tubes into the water saturated gravel above the tunnel. Altogether, 1'100 m^3 cement suspension with 4.4 tons of technical SNFC mixture were applied in order to stabilize the soil prior to construction and to minimize water intrusion into the tunnel. Three piezometers (No. 1, 2, 3) situated at a horizontal distance of about 5-10 m from the future tunnel allowed sampling of the groundwater in a depth of 48 to 57 m. The groundwater is characterized by the following parameters: T ≈ 12°C, pH 7.5, 11 mg/L O_2, 0.3 mg/L DOC (dissolved organic carbon), 90 mg/L Ca^{2+}, 3.5 mg/L Cl^- and 14 mg/L NO_3^-. Groundwater samples from the construction site were collected in amber glass bottles and stabilized on site by adding 1 % (v/v) formaldehyde solution (37 % w/w). The samples were stored at 4°C. Sludge from piezometer No. 1 was used for the biodegradation experiments and was also stored at 4°C until incubation.

Field site B (Zurich-Thalwil) is a tunnel construction site, where injections into the aquifer with cement suspensions containing SNFC were made. The site is located in an aquifer with an average groundwater flow velocity of 1 m/d. The sampled piezometer was about 100 m downstream of the construction site.
For reference biodegradation experiments, activated sewage sludge was taken from the aeration basins of two Swiss wastewater treatment plants (WWTP). Both WWTP received wastewater from textile finishing industry, where sulfonated naphthalenes were applied.

3.2.3 Instruments

SNFC was determined with a liquid chromatograph (HPLC) 1090L Series II coupled to a diode array detector (UV) and a fluorescence detector (FLD) HP 1046 (Hewlett Packard Schweiz AG Urdorf, Switzerland).

3.2.4 Biodegradation Experiments

The experiments were conducted in 250-mL clear bottles with screw-caps. The growth medium contained 20 mM phosphate buffer (pH 7, KH_2PO_4 and Na_2HPO_4), 0.1 mM S ($MgSO_4$), 0.1 mM N (NH_4Cl), 0.25 mM Ca ($CaCl_2$), and trace elements (Fe, Mn, Zn, Cu, Co, Mo and Ni). The inoculum (15 mL of piezometer sludge or WWTP sludge) was added to 135 mL of growth medium. The test substances (technical SNFC mixture, N15S, N16S, N17S, N26S, or N27S) were added in amounts of 38 to 70 mg/L each as only C-source. Five mL of formaldehyde solution (37 % w/w) was added to a control experiment with piezometer sludge and technical SNFC mixture, in order to inhibit bacterial growth (negative control). The incubations were stored in the dark at 25°C and aerated by shaking. The first samples were taken 3 h after the start of the experiments and then samples were collected every two or three days. The samples were frozen and stored at -18°C until analysis. Before measurement the samples were filtered through a 0.2 µm membrane filter (Teflon, Schleicher und Schuell, Riehen, Switzerland) in order to remove bacteria.

In addition, reference samples were filtered in order to compensate for adsorption of SNFC components onto the filter.
The lag phase was defined as the time interval from the start of the incubation to the time point at which the substrate concentration decreased to 90 %. The maximum degradation rate based on zero order kinetics was obtained from the slope of the linear part of the degradation curve. The degradation phase was defined as the time period from the end of the lag phase to the end of the degradation (substrate concentration < 1 %).

3.2.5 Liquid Chromatography with UV and Fluorescence Detection

Environmental samples were prepared and analyzed according to the procedure published by Wolf et al. [14]. Briefly (see also [15]), 500 mL environmental samples, with 1 mmol/L TBABr and 2.5 µg/L of internal standard (diphenylamine-4-sulfonate), were percolated through a cartridge with 1 g of solid-phase material (LiChrolute) for about 1 h. The cartridges were eluted with 5 mL acetonitrile. The solvent was evaporated with nitrogen, then the dry residues were redissolved in 500 µL of a 4 mM TBABr aqueous solution. The samples from the biodegradation experiments were diluted 50 fold by adding 4 mM TBABr aqueous solution. 50 µL of sample were injected onto the HPLC column (octadecylsilica column Hypersil ODS, 5 µm, 250 x 4 mm i.d. with a precolumn 5 x 4 mm i.d., Macherey-Nagel AG, Switzerland). The aqueous mobile phase (A) was HPLC-grade water with 4 mM TBABr. A 25/75 mixture of water and acetonitrile with 4 mM TBABr was used as organic modifier (B). Starting with 30 % B, a gradient was run over 40 min to 35 % B, over an additional 40 min B, was increased to 80 % and finally run for another 5 min to 100 %. 10 min were run isocratically with 100 % B. The HPLC was brought back to initial conditions over a period of 5 min, followed by 2 min of equilibration. The flow rate was 0.8 mL/min and the column temperature was held at 40°C. The UV-absorption wavelength for the internal standard was 300 nm, the FLD excitation wavelength was 230 nm and the acquired emission wavelength was 360 nm. The photomultiplier gain was 9. The substances were identified based on their absorption and emission wavelengths and their retention times in the FLD chromatogram. The concentrations of the individual compounds were calculated by external calibration. The calibration of the dimer was used for all oligomers, since the degree of polymerization has no influence on the response factor [14]. Each environmental sample was enriched and analyzed twice. The recoveries were between 59 and 104 %, depending on the compounds, the relative standard deviations were between 3 and 11 %, the limits of detection between 0.01 and 0.04 µg/L and the limits of quantification between 0.03 and 0.15 µg/L. The relative standard deviations of the samples from the biodegradation laboratory experiments were between 4 and 6 % (n=5).

3.3 Results

3.3.1 SNFC in the Groundwater of the Field Sites

The construction activities at field site A influenced the aquifer as follows: No SNFC was found in the groundwater before the injection activities started. Two weeks after the first injections, the total SNFC concentration in piezometer No. 1 (KB 2) rose to a maximum concentration of 230 µg/L. Afterwards, SNFC concentrations decreased and further injections did not significantly influence SNFC concentrations in the groundwater anymore.
The analysis of the groundwater samples taken at piezometer No. 2 (KB 4) during the injection activities showed that all monomers of the technical SNFC mixture were found at the same relative concentrations as in the technical SNFC mixture. However, the relative concentrations of the oligomers in the groundwater samples were lower than in the SNFC mixture (Figure 3.2, Trace A and B) and no oligomers with chainlengths of more than four units were detected. Still, the isomeric distribution pattern was found to be similar to the technical SNFC mixture. After the end of the injection activities, the concentrations of some monomeric SNFC components decreased very quickly. This led to a changing distribution pattern of the components, which can be seen in the HPLC-FLD chromatograms in Figure 3.2, Trace B and C. When observing the concentrations of the individual SNFC compounds over the observation period, it can be seen that the first components to disappear were N1S and

N2S (Figure 3.3). Twenty days later, N16S and N26S could no longer be detected and finally N27S and N17S were eliminated after 80 and 200 days, respectively. The only monomeric component to be found over the whole monitoring time was N15S. It showed an almost linear concentration decrease. Also the concentration of the oligomers decreased linearly and the isomeric distribution did not change.

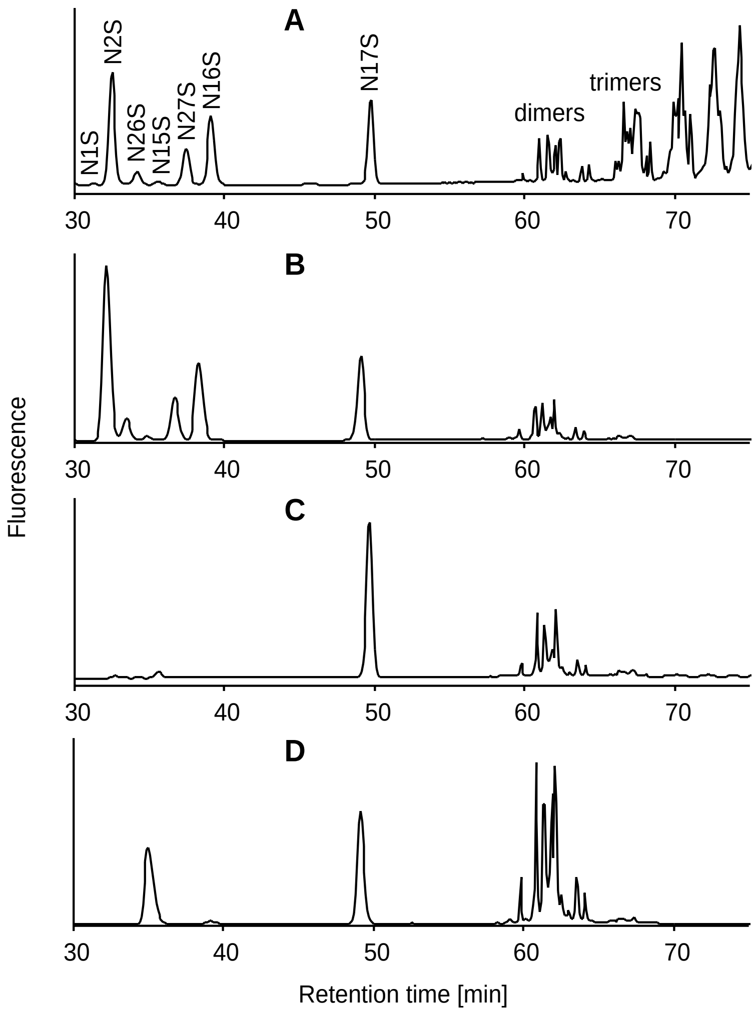

Figure 3.2: *HPLC-FLD chromatograms of the technical SNFC mixture (A), the groundwater of field site A from piezometer No. 2 at the beginning of the injection activities (B), the groundwater from piezometer No. 2 100 d after the end of the injections (C), and the groundwater from the piezometer of field site B, where the flow time from the construction site to the piezometer was about 90 days (D).*

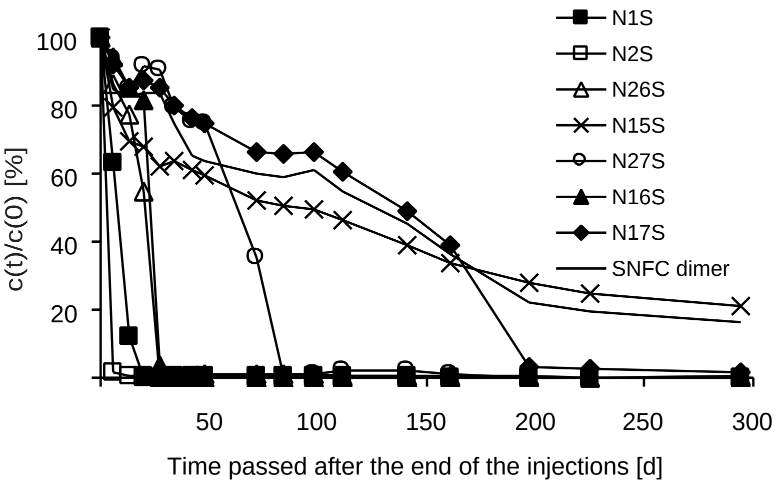

Figure 3.3: *Behavior of the sulfonated naphthalene monomers and the dimer (as a representative for the SNFC oligomers) in the aquifer of field site A at piezometer No. 2 (KB 4). The concentrations (c(t)) were normalized to the concentration of each component at the end of the injection activities (c(0)).*

Additionally, groundwater was analyzed at field site B where construction activities in the aquifer were performed. The sampling point of field site B was located 100 m downstream of the construction site. Hence, with a groundwater flow velocity of about 1 m/d the SNFC plume needs about 3 months to reach the piezometer. In the groundwater of the sampling point of field site B only N15S, N17S and the dimer could be detected (Figure 3.2, Trace D). After the end of the injection activities at field site B the ratio between N17S and N15S decreased and 4 months later N17S was not detected.

3.3.2 Biodegradation Experiments

The technical SNFC mixture in the growth medium was incubated with sludge from piezometer No. 1 of the field site A at 25°C. Concentrations of sulfate and oxygen and the pH were similar to the ones in the groundwater of the field site. In the negative control experiment, the concentrations of the different components of the technical SNFC mixture remained constant during the course of the whole experiment. The results of the degradation experiment with piezometer sludge are presented in Figure 3.4 A and B and Table 3.1 (row A). N2S, N1S, N16S, N26S and N17S were degraded after different lag phases and with different maximum degradation rates. N27S, N15S and the oligomers were not degraded within 195 d. The degradation of the monomeric components was found to follow zero order kinetics.

Individual disulfonated monomers were incubated separately with aquifer material and growth medium in an additional set of experiments. The results are shown in Figure 3.4 C and Table 3.1, row B. The resulting degradation parameters were similar to those obtained with the technical mixture. Only the maximum degradation rates of N16S and especially N26S differed: The degradation of N26S as sole substrate was found to be 10 times more rapid. Still, the lag phases for the degradation of N16S and N26S were about the same.

To check the adaptability of the sludge, the technical SNFC mixture was incubated in growth medium with piezometer sludge from previous SNFC degradation experiments. This sludge was generally able to degrade the monomeric disulfonated naphthalenes faster than unadapted sludge (Table 3.1, row C). In contrast to the other experiments, in which N27S was not degraded it was degraded in this experiment. N16S and N26S were degraded almost simultaneously (see lag and degradation phase) as seen in previous experiments (Figure 3.4 B). Again, N15S and the oligomeric SNFC were not degraded.

Additional degradation experiments were made with activated sludge from two Swiss wastewater treatment plants (WWTP) receiving wastewater from textile finishing industries (Table 3.1, row D and E). These sludges were adapted to SNFC due to the application of SNFC as dispersant, wetting and tanning agent in this particular industry. However, even in such experiments the oligomers were not degraded, although the sludges were able to degrade all disulfonated monomers with the exception of N15S.

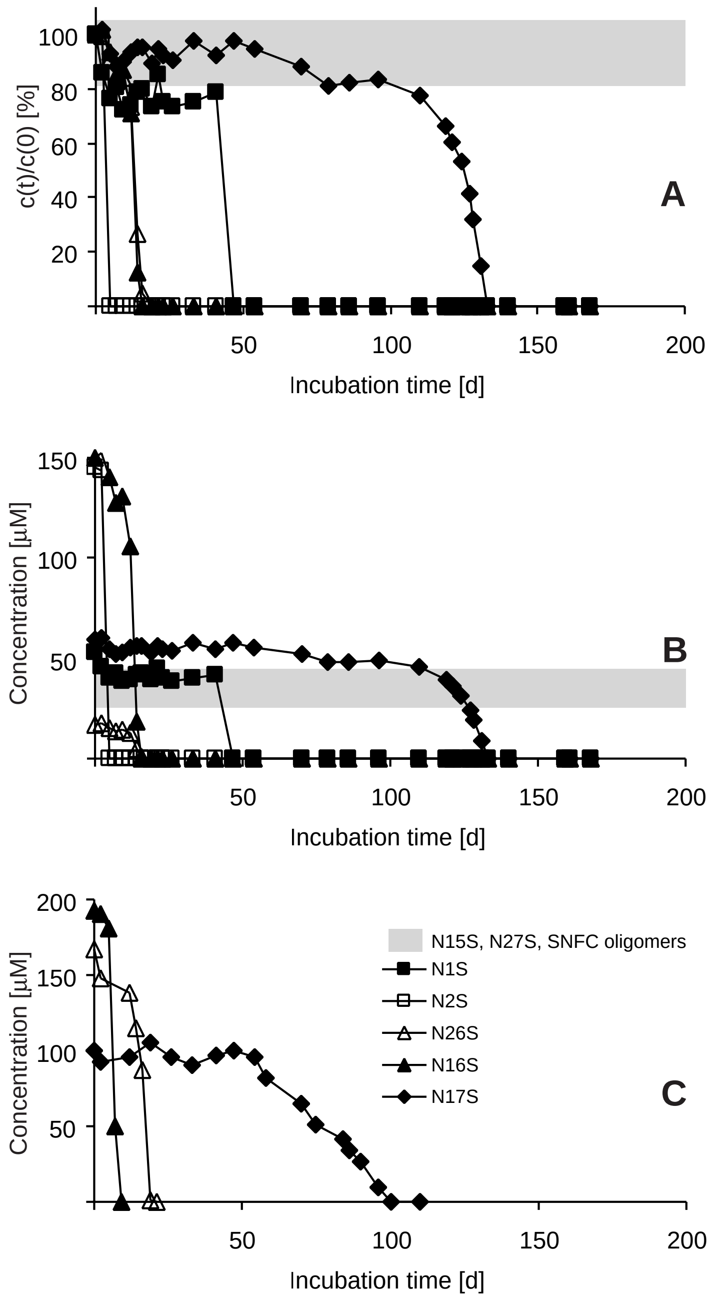

Figure 3.4: *Aerobic biodegradation of the technical SNFC mixture incubated with piezometer sludge as inoculum. A: The concentrations (c(t)) were normalized to the concentration of each component measured after 3 hours of incubation (c(0)). B: The concentrations were not normalized [µM]. C: Degradation of the disulfonated naphthalenes incubated separately with piezometer sludge. The concentrations of persistent SNFC components are within the gray bands. In the control experiments, the concentrations of all components remained constant.*

Table 3.1: *Lag Phases, Maximum Degradation Rates, and Degradation Phases for the Degradation of the Different SNFC Components under Varying Conditions, A: Technical SNFC Mixture Incubated with Piezometer Sludge, B: Disulfonated Monomers Incubated Separately with Piezometer Sludge, C: Technical SNFC Mixture Incubated with Adapted Piezometer Sludge, D + E: Technical SNFC Mixture Incubated with Sludge from Two Different Wastewater Treatment Plants*

	N1S			**N2S**			**N26S**			**N15S**			**N27S**			**N16S**			**N17S**			**oligomers**		
	lag	*rate*	*deg*	*lag*	*rate*	*deg*	*lag*	*rate*	*deg*	*lag*	*rate*	*deg*	*lag*	*rate*	*deg*	*lag*	*rate*	*deg*	*lag*	*rate*	*deg*	*lag*	*rate*	*deg*
A	0-2	7	47	2-5	48	3	2-5	3	17	>195	-	-	>195	-	-	5-7	20	14	54-70	3	86	>195	-	-
B							0-2	29	21	>195	-	-	>195	-	-	5-7	45	7	54-58	3	46			
C	2-5*	17	19	2-5	46	5	2-5	7	7	>195	-	-	2-5	6	17	2-5	28	10	86-96*	4	n.a.	>195	-	-
D	12-14	12	21	0-2*	20	14	33-41	4	45	>195	-	-	41-43	7	47	44-47	30	49	23-26*	17	76	>195	-	-
E	0-2	19	7	0-2	27	5	2-5	3	12	>195	-	-	33-35	1	143	2-5	25	12	7-9*	24	33	>195	-	-

lag = lag phase [d], rate = maximum degradation rate [µM/d], deg = degradation phase [d]. The * indicates a degradation in two steps. The ranges for the lag phases could not be determined more precisely due to the sampling intervals of some days. n.a. = not achieved during the time of the experiment.

3.4 Discussion

3.4.1 SNFC in the Groundwater of the Field Sites

The groundwater analysis of field site A showed that the injection activities in the groundwater saturated gravel had a direct influence on the concentration of SNFC in the groundwater. The SNFC concentrations rose two weeks after the start of the injections. This time lag most likely reflected the time SNFC needed to reach the piezometer. Further injections did not influence the groundwater significantly, which can be explained by reduced contact of fresh cement with groundwater due to cement barriers from the earlier injections.
The distribution of the monomers in the groundwater during the injections reflected the distribution of the injected technical SNFC mixture. This gives evidence that all monomers had identical leaching behavior and that the groundwater was not yet influenced by biological activity. After the end of the injection activities some monomeric SNFC components showed a concentration decrease which exceeded the linear concentration decrease of N15S and the oligomers. Since the linear decrease was observed for the persistent components (see results of laboratory experiments) this decrease was most likely due to the dispersion of the SNFC plume in the stagnant body of groundwater. The enhanced decrease of the other monomers was interpreted as biodegradation. This interpretation was confirmed in the laboratory experiments.
In the case of the oligomers, the distribution pattern during the injection activities was not identical to the one of the technical mixture. Since this observation could not be explained by biological activity in the aquifer (no degradation of oligomers was observed in the laboratory experiments) it must be concluded that the oligomers had different leaching characteristics from the monomers, due to different adsorption properties. This view was also suggested elsewhere [14, 16, 17] and can be explained with ionic interaction between the anionic sulfonate groups and the positively charged surface of the cement. Because the number of reactive groups per molecule increases with the degree of polymerization, adsorption onto the cement becomes more and more relevant for higher oligomers. The effect is predominant for oligomers with more than four units, which explains why

these oligomers could not be detected in the groundwater samples of field site A. The leached oligomers were not biodegraded (Figure 3.3). Their concentration decreased simultaneously with the concentration of the persistent N15S indicating that the main process affecting their fate was dispersion. This view was substantiated by the fact that the isomeric distribution of the oligomers did not change over the entire observation period. In case of biodegradation, the distribution pattern of the oligomers should change because the lag phases and kinetics for the different oligomers are expected to be distinct. Additionally, unidentified signals in the chromatogram indicating the production of degradation intermediates from the oligomers were not observed.

The presence of biological activity in the aquifer also explains the results obtained at field site B. During time period the groundwater needed to reach the sampling point of field site B, all components except N15S, N17S and the oligomers were degraded. Since N15S, N17S and the oligomers were not degraded within 3 months at field site A (Figure 3.3), it is reasonable that they were the only components to be found in the groundwater of field site B. However, the changed ratio of N15S to N17S when compared to the ratio in the technical SNFC mixture (Figure 3.2, Trace D) indicates that N17S was biodegraded in the aquifer of field site B. Effectively, the N17S concentration decreased relative to the N15S concentration during the observation period, indicating adaptation of the micro-organisms in the aquifer. Four months after the end of the injections, the concentration of N17S was below the detection limit.

3.4.2 Biodegradation Experiments

The biodegradation experiments with piezometer sludge showed that the monosulfonated monomers were degraded more quickly than the disulfonated analogues. This is in accordance with the literature [10, 18]. In the laboratory experiments, the oligomers persisted. Questions regarding their degradability under different environmental conditions appeared interesting, since the degradation of oligomers could be expected in analogy to the degradation of the disulfonated monomers. But the oligomers were persistent even when incubated with adapted populations of micro-organisms (see Table 3.1, row C-E). Experiments, in which disulfonated monomers were separately incubated showed that the degradation of one particular component of the SNFC mixture is not affected by the presence of the other components. Additionally, it was confirmed that neither N27S nor N15S could be degraded even if they were the sole substrates.

The linear degradation of the monomers found in the laboratory experiments indicated zero order degradation kinetics. From this we conclude that the substrate was in excess and that it did not influence the degradation rate. Assuming enzymatic degradation, the limiting factor for the degradation most probably was the amount of active enzyme.

The degradation pattern for the single components of the technical SNFC mixture in batch experiments was similar to the one observed at the field site (Figure 3.3 and 3.4 A). In both cases the monosulfonated monomers were degraded first (in the case of N1S in the laboratory experiments two steps were observed) and the disulfonated monomers N16S and N26S were degraded simultaneously. N17S was degraded only after more than hundred days and N15S and the oligomers were persistent in the laboratory experiment and at the field site. The only difference concerns N27S which was not degraded in the experiments with piezometer sludge, but was degraded in all other cases.

In conclusion, the results of the laboratory degradation experiments and the results of the field measurements strongly support the notion that the most important process for the elimination of SNFC components at the field site was biodegradation. The experiments also show that the degradation sequence of the different monomers was variable and depended on environmental conditions and on the presence of adapted micro-organisms. The experiments revealed that the monosulfonated monomers were readily degraded, while the disulfonated analogues were not as easily metabolized. N15S and the oligomers proved to be persistent. The degradation kinetics were found to be zero order. The presence of many different SNFC components did not affect the biodegradability of the single component.

This study shows that SNFC used as superplasticizers for cement suspension injections in the groundwater saturated gravel leached into the groundwater. Our results show that although most monomers were degraded, some of the components, namely N15S and the oligomers, were persistent in the aquifer.

Acknowledgments

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4

Sorption and Mass Flux Calculations of SNFC

Sulfonated naphthalene formaldehyde condensates (SNFC) and their monomeric analogues were used as superplasticizers for cement suspension injections at two tunnel construction sites that are in direct contact with groundwater. Because in one case the aquifer is an important drinking water resource, the behavior of SNFC in the groundwater was carefully investigated. Chemical analyses showed that SNFC leached to the groundwater in concentrations of up to 58 μg/L of total SNFC at a distance of about 55 m downgradient from the construction site. Of the individual SNFC components, only monomers and oligomers with up to four units could be detected in the groundwater. Oligomers with more than four units did not leach from the cement paste. The leached oligomers were transported in the groundwater at different velocities, which can be explained by sorption experiments. Mass fluxes of SNFC used at the tunnel construction sites were evaluated. Most SNFC were immobilized in the cement, but 5 % (w/w) of the applied SNFC were found to leach into the aquifer. This corresponds to a total amount of leached SNFC of approximately 100 kg, of which about 80 % are biodegraded in the aquifer and 20 % (20 kg) still remain in the groundwater.

Ruckstuhl, S., Suter M. J.-F., Giger, W. *Sorption and Mass Fluxes of Sulfonated Naphthalene Formaldehyde Condensates in Aquifers.* Submitted to Water Research.

4.1 Introduction

Modern concrete construction technology is increasingly based on chemical admixtures, which allows control of the concrete properties. These chemicals are mostly organic, water-soluble compounds with high production volumes (in Switzerland, e.g. in the range of several hundred tons per year). Their toxicity is variable, depending on the constituents of the product. When used in aquifers for underground work, they are expected to leach and contaminate the aquatic environment. In fact, the application of construction chemicals at two tunnel construction sites in northern Europe caused severe environmental problems. In both cases, the inflow of groundwater into the tunnel was stopped with injections of toxic grouting agents based on acrylamide in combination with cement. Symptoms characteristic for exposure to the grouting agent were observed for workers in the tunnels. In one of the two cases the leached grouting agent exfiltrated to the surface water. Poisoning of fish and cattle was observed and several water wells were contaminated [1-3].

Construction chemicals are also applied in Switzerland in highly permeable alluvial aquifers that are supplies for drinking water. However, to the best of our knowledge no field-scale studies on the leachability of organic construction chemicals have been performed. Only one recent publication reports on the transport behavior of one representative of construction chemicals in a groundwater system [4]. Thus, this study focuses on the leachability and the environmental fate of one group of representative construction chemicals, namely sulfonated naphthalene formaldehyde condensates (SNFC). The mass fluxes in the groundwater at two tunnel construction sites, where cement suspensions were injected in tunnel surroundings to stabilize the gravel and sand for the drilling work were examined. SNFC are the most important concrete admixtures, accounting for 10 % of the total applied concrete admixtures in Switzerland in 1992. The world-wide annual consumption is 150'000 t [5]. The technical SNFC mixture consists of several mono- and disulfonated monomers and their condensed oligomers with up to about 15 to 20 units (Figure 4.1). SNFC have low octanol-water partition coefficients (log $K_{ow} = -0.94$), low pK_a values (< 1) [6, 7] and, in contrast to the construction chemicals applied in northern Europe, a low toxicity (e.g. fish toxicity: LC_{50} = 100 - 500 mg/L) [8]. The less condensed SNFC oligomers were shown to leach from fresh cement when in contact with groundwater. In groundwater, naphthalene-1,5-disulfonate and the SNFC oligomers were found to be persistent [9]. Possible environmental effects are contamination of drinking water resources and for the oligomers to remobilize nonpolar organic contaminants, due to their properties as surface active agents. Thus, the Swiss authorities established a provisional standard for SNFC of 1 mg/L at one of the investigated tunnel construction sites.

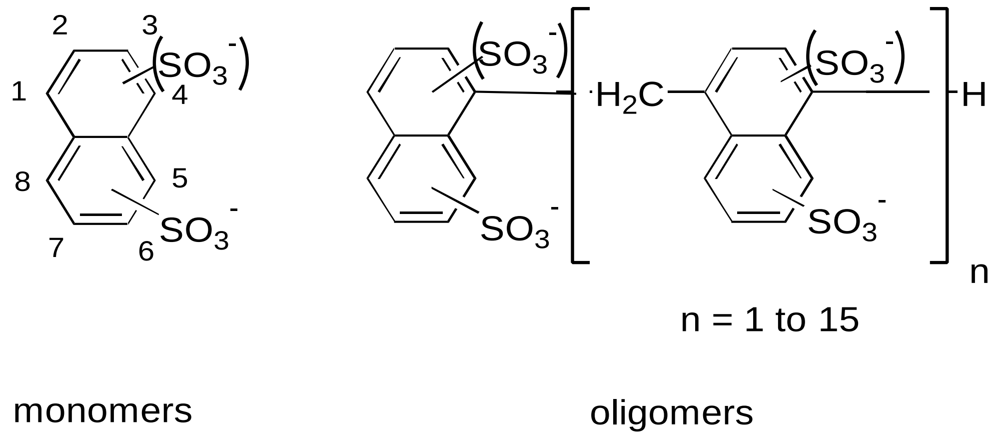

Figure 4.1: *Chemical structures of the components of technical SNFC mixtures: Monosulfonated and disulfonated naphthalenes (monomers) and formaldehyde condensates (oligomers). All three compound classes consist of various isomers.*

This study aimed to understand the fate of SNFC in the groundwater. Therefore, sorption phenomena were studied by analyzing groundwater from the affected aquifer and by performing laboratory experiments. From the measured SNFC concentrations in the groundwater, the total amount of SNFC leached was estimated. This data, combined with information on biodegradability, allowed the calculation of the overall SNFC mass fluxes at the tunnel construction sites in contact with groundwater and provides basic information for assessing the environmental risk of polar organic construction chemicals.

4.2 Materials and Methods

4.2.1 Chemicals

Naphthalene-1-/-2-sulfonate (N1S, N2S), naphthalene-1,5-/-1,6-/-2,6-/-2,7-disulfonate (N15S, N16S, N26S, N27S) and diphenylamine-4-sulfonate were obtained from Fluka AG (Buchs, Switzerland), Aldrich (Steinheim, Germany) and TCI (Tokyo, Japan). Naphthalene-1,7-disulfonate (N17S) and the condensed naphthalene-2-sulfonate dimer (8,8'-methylenebis-2-naphthalenesulfonate) were kindly made available by TZW Karlsruhe, Germany [10]. "Technical SNFC mixture" refers to Galoryl LH 120 from CFPI Industries (Gennevilliers, France). The technical SNFC mixture contains about 12 % SNFC, of which 11 % were monosulfonated monomers, 23 % were disulfonated monomers and 66 % were oligomers.

4.2.2 Tunnel Construction Sites

The Zurich tunnel construction site is situated over a length of about 400 m in an unsaturated gravel-and-sand aquifer. The direction of the tunnel is oblique to the groundwater flow direction. The gravel had to be stabilized prior to drilling. A cement suspension with SNFC (8 kg/m^3 technical SNFC mixture) as a superplasticizer was injected from a pilot tunnel into the partly water saturated gravel above the future tunnel. The injections were performed through drilled tubes with pressure sleeves.

Twenty three piezometers were placed at a distance of 25 to 130 m downstream of the construction site at different sections along the tunnel, monitoring for groundwater quality over a period of two years. In the monitoring program, pH, conductivity, dissolved organic carbon (DOC), silica, Na^+, K^+, Cr^{VI} and SNFC were measured regularly in the aerobic groundwater at different sections along the tunnel construction site and at different distances from it. If the concentration of one parameter had exceeded the standard at a distance of more than 100 m from the construction site, the contaminated groundwater plume would have been pumped out of the aquifer through five decontamination wells each of a

capacity of 1500 L/min. The contaminated water would then have been disposed into a municipal wastewater treatment plant.
From the piezometers, groundwater was pumped for sampling with a positive displacement pump (discharge rate 28 L/min), and 450 L were pumped from each piezometer before sampling. All samples were collected in amber glass bottles and stored at 4°C.
Another tunnel construction site was studies at Baregg. A cement suspension with 4 kg/m^3 technical SNFC mixture was injected into the aquifer. Three piezometers, which are situated at a horizontal distance of about 5-10 m from the tunnel, allowed sampling of the almost stagnant groundwater.

4.2.3 Sorption Experiments

Batch experiments were carried out with aquifer material taken from a depth of about 2 m. 2.5 g of aquifer material were added to 5 mL of groundwater. The aquifer material was dried at 110°C over night and then sieved. The fraction with a diameter of 0.16 to 0.50 mm was used for the experiments. The surfaces of this fraction were assumed to be representative for the flowing groundwater (without pore water). The groundwater used is of a chemical composition comparable to the one at the field site (specific electric conductivity 670 µS/cm, pH 7.4). Technical SNFC solution was added in concentrations between 0.4 and 400 mg/L (for the sorption kinetics measurements: 4 mg/L). The tests were conducted in 25-mL clear vials with screw-cap. The vials were shaken overhead with 22 rounds/min at 25°C. The gentle motion was assumed to have negligible effect on the surfaces of the aquifer material. No measures against biodegradation were taken, since N15S and the oligomers were found to be persistent [9]. For the kinetic experiments, samples were taken between 10 min and 57 days, and for the sorption isotherm experiments after 33 days. The sorption process was stopped by filtering the experimental mixture with a centrifugal device (0.45 µm, Ultrafree-MC, Millipore, Bedford, MA). The oligomers with a chainlength of two to six sorb between 2 and 18 % to this type of filter. Hence, reference samples (without aquifer material) were also filtered to account for the sorption onto the filter material. The pH of the samples rose slightly during the experiment (from 7.4 to 7.8 within 30 d). The amount of SNFC remaining in the water phase after the sorption process was measured with HPLC-FLD. The sorbed amount was calculated from the difference to the reference samples.

4.2.4 Analysis (HPLC-FLD)

The samples from the sorption experiments were diluted up to hundred times and groundwater samples were enriched before measuring. The enrichment and measurements were performed according to the procedure developed by Wolf et al. [11] and modified by Ruckstuhl et al. [12]. The recoveries for groundwater samples were, depending on the compounds, between 59 and 104 %, with a relative standard deviations between 3 and 11 %, the limits of detection between 0.01 and 0.04 µg/L and the limits of quantification between 0.03 and 0.15 µg/L. The relative standard deviations for the analysis of the samples from the sorption experiments at one concentration were between 4 and 13 % (n=6).

4.3 Results and Discussion

4.3.1 SNFC in the Groundwater of the Zurich Construction Site

Before the start of the injections no SNFC could be detected in the groundwater. Traces of SNFC were observed about one month after the start of the injection activities at the downgradient piezometer 1 (KB ZT 91-6). This piezometer is located 25 - 30 m downgradient from the section of the tunnel, in which the construction in the aquifer started. Thus, the fastest components of the SNFC plume, i.e. the monomers, moved at a velocity of about 1 m/d, which is the estimated average flow velocity of the groundwater in this area (estimation based on tracer tests with fluorescein and naphthionate). A second piezometer (2, KB ZT 91-4) is located at a distance of about 55 m downgradient from a section of the construction site which is different from that of piezometer 1. In this area the groundwater flows with an average velocity of 6.5 m/d. At piezometer 2, the first SNFC components detected were monomers and dimers. But with increasing observation time also trimers and later tetramers appeared in this piezometer (see Figure 4.2).

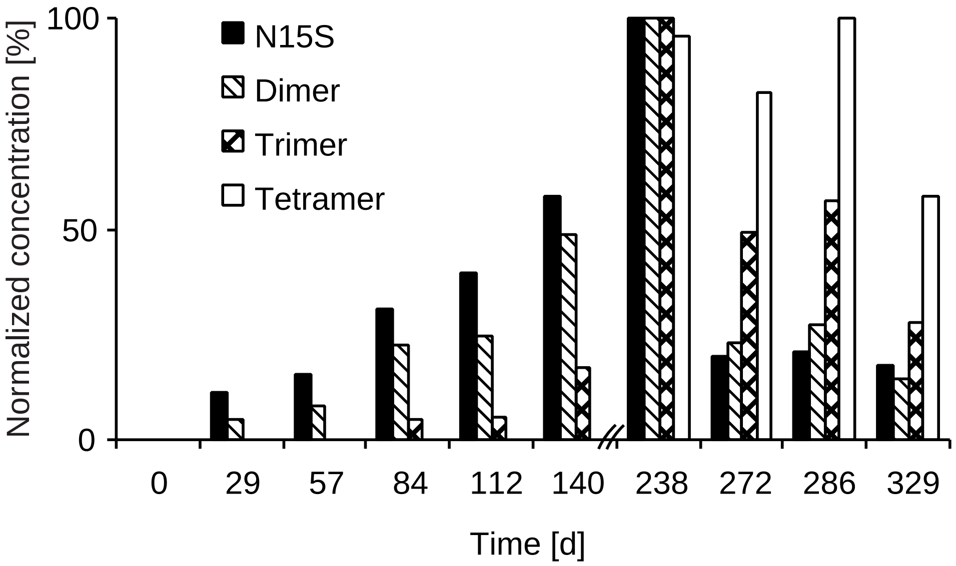

Figure 4.2: *Sequential appearance of a disulfonated naphthalene (N15S) and different SNFC oligomers at a groundwater sampling point 55 m downstream of the tunnel construction site. The concentrations are normalized to the peak concentration observed. Maximum concentrations were 35.4, 17.7, 3.5 and 0.43 µg/L for N15S, dimer, trimer and tetramer, respectively.*

SNFC leaching from the injected cement suspension were also detected at several other piezometers. The highest SNFC concentration found in the groundwater was 58 µg/L in piezometer 2. This concentration is still well below the standard for SNFC issued by the Swiss authorities. Oligomers of more than four units were detected neither in piezometer 2 nor in any other of the monitored piezometers over the whole observation time of two years.

4.3.2 Sorption Experiments in the Laboratory

In order to explain the observations of the Zurich field site sorption experiments were undertaken. Figure 4.3 shows the sorption kinetics of SNFC and the monomeric sulfonated naphthalenes in these experiments. N15S as a persistent representative for the monomers and the dimers sorbed very slowly onto the aquifer material. N15S and the dimers reach equilibrium after 30 days with a maximum sorption of about 45 %. The oligomers with a chainlength of more than four units were sorbing onto the aquifer material within hours to an extent of more than 90 %. The trimers and the tetramers reached equilibrium after about 30 days with a maximum sorption of 86 and 96 %, respectively.

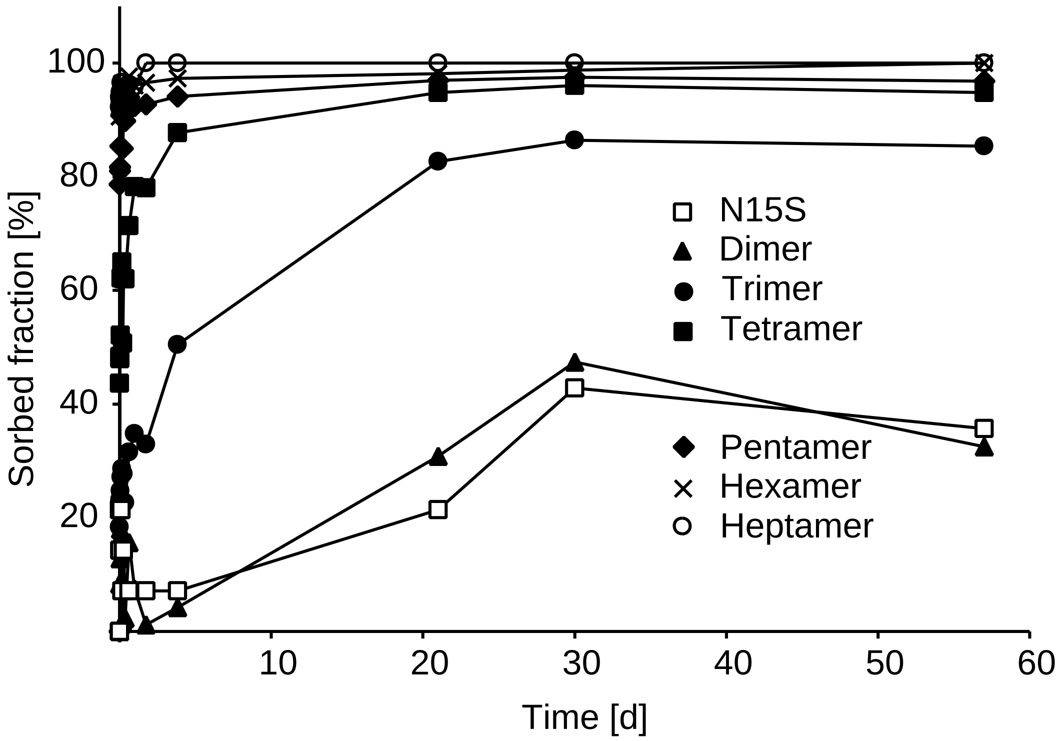

Figure 4.3: *Sorption kinetics of N15S (as a representative for the monomers) and of oligomeric SNFC. Solid phase: Natural sand, 0.16 - 0.5 mm diameter, dry sieved. The absolute initial concentrations were 0.12, 0.14, 0.23, 0.26, 0.28, 0.23 and 0.22 mg/L for N15S, dimer, trimer, tetramer, pentamer, hexamer and the heptamer, respectively.*

In Figure 4.4, the sorption isotherms of the oligomers are shown. The isotherms are steep at the beginning, which expresses the high affinity of the components to the aquifer material surface. The higher condensed oligomers have steeper slopes confirming a better sorption behavior of these components. The subsequent flattening of the isotherm occurs at concentrations that are far above those measured in the aquifer. This indicates saturation of the sorption sites of the aquifer material surface, leading to an increasing concentration of the solved components. From the slope of the initial linear section of the isotherms, volume based solid-liquid partitioning ratios, R_d, were obtained. Values of 200, 90, 20, 2 and 1 mL/g were calculated for the pentamer, tetramer, trimer, dimer and monomer (N15S), respectively.

In order to compare the findings of the sorption experiments with the field observations, retardation factors, R_f, of the solutes with respect to the flowing groundwater were calculated as $R_f = 1 + R_d*\rho(1-\varepsilon)/\varepsilon$, with ρ = density of the aquifer material and ε = total porosity. The R_f-factors are based on the experimental R_d-values and aquifer properties of an aquifer in the same geological area [13]. The retardation factors calculated for the laboratory system (R_f of 20 - 900 for the dimers to the trimers) were found to be higher than these in the field by one or two orders of magnitude. This can be explained by the very heterogeneous character of the surface sites of aquifer materials, the differences of experimental scales, the different SNFC concentrations in the two systems, the differences of the solid-liquid-phase-ratio and the aquifer material fraction selected for the laboratory experiments.

Qualitatively, the similar R_d-values of N15S and the dimers in the experiments could explain their nearly simultaneous occurrence at the piezometer (the sampling interval in the field of one months did not allow for a more precise interpretation) and the increase of the experimental R_d-values with the chainlength of the oligomers could explain their decreasing transport velocity in the aquifer.

The mechanism responsible for the sorption in the aquifer is most probably based on ionic interactions between the negatively charged sulfonate groups and the partly positively charged surfaces of aquifer components like Fe(III)(hydr)oxides. Because the number of sulfonate groups per SNFC molecule increases with the degree of polymerization, sorption is progressively stronger for the higher oligomers. Sorption by hydrophobic interaction is unlikely, since the hydrophobicity remains about the same with increasing degree of condensation [11].

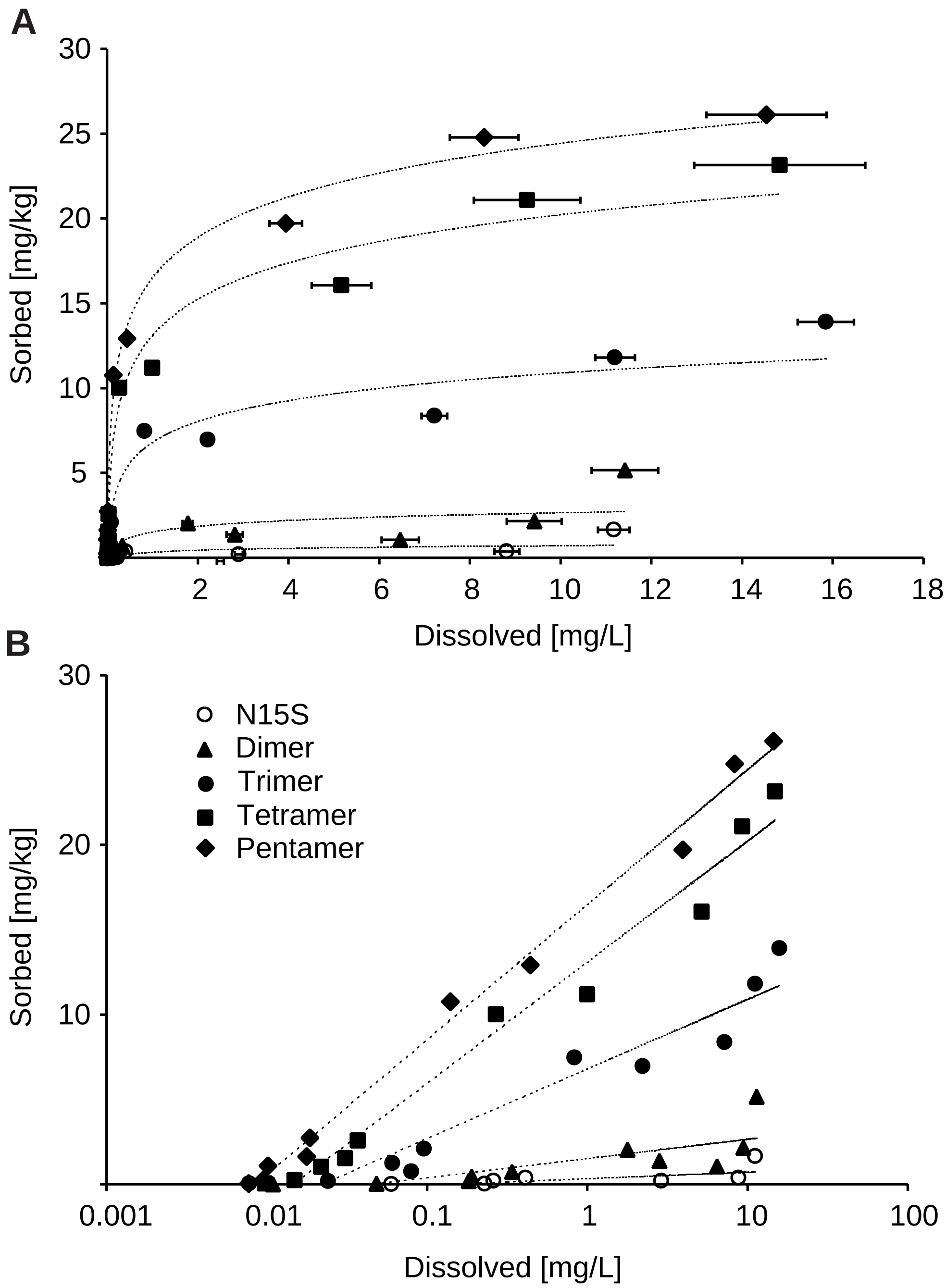

Figure 4.4: *Sorption isotherms for N15S and SNFC oligomers onto natural sand (0.16 - 0.5 mm diameter; dry sieved). Lower graph with logarithmic x-axis.*

This sorption mechanism is also responsible for the leaching behavior of SNFC from cement [9]. The oligomers with more than four units sorb immediately onto positively charged surfaces of the cement [14], and after the hardening of the cement, they are immobilized in the cement matrix [15].

4.3.3 Mass Fluxes at the Construction Sites

Mass fluxes of SNFC were calculated for the Zurich construction site in order to achieve an overall assessment of the environmental fate of SNFC. Source, pathways and sinks are shown in Figure 4.5 and are discussed below. The system boundary of 195 days was chosen because the persistent SNFC components were defined as not biodegraded within 195 days [9].

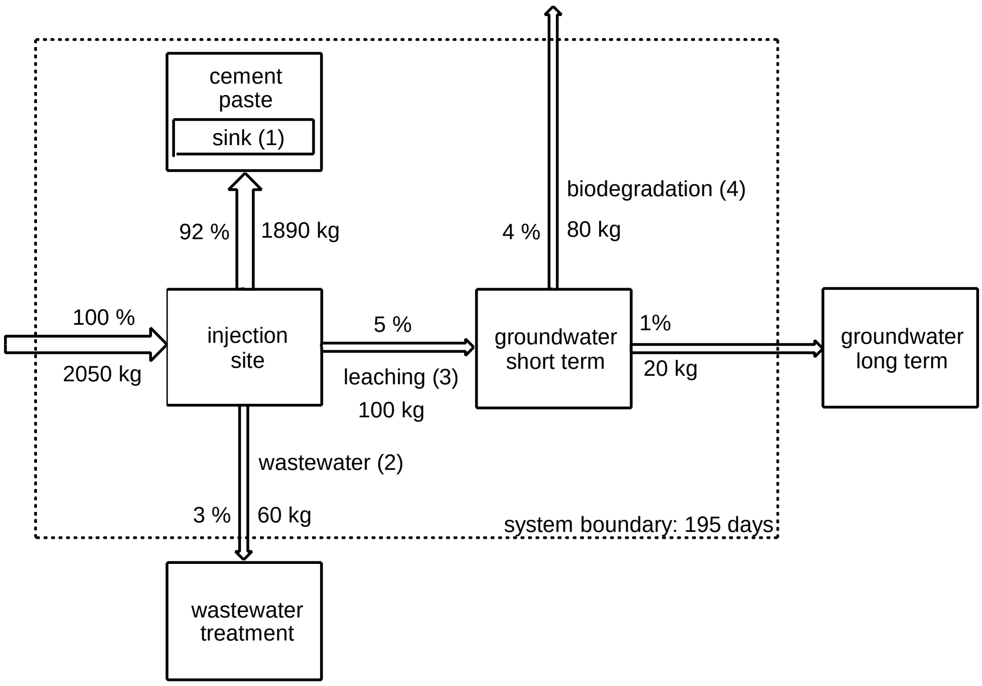

Figure 4.5: *Mass fluxes for SNFC at the Zurich tunnel construction site in percentages (related to the 100 % of input) and in absolute amounts. The numbers in parenthesis refer to the sink description in the text. Thicknesses of arrows are not in scale.*

Source of SNFC: The only source of SNFC is the technical SNFC mixture, which was used for the injections. Over the injection period of 1.5 years 16.7 tons of technical SNFC product, an aqueous solution with a content of 12.3 % of SNFC were used, accounting for a total of 2050 kg SNFC.

Sinks of SNFC: **(1)** The major sink of SNFC was the hardened cement, in which the bulk of the SNFC was immobilized. The mass flow of 1890 kg SNFC (Figure 4.5) was calculated by subtracting the output through the wastewater and through leaching from the input.

(2) Part of the SNFC was supposed to be lost at the cement suspension mixing place and inside of the pilot tunnel (injection suspension flowing back from the drilled tubes before the hardening of the cement). This part of the SNFC was collected with the wastewater and led into the pretreatment plant of the construction site, where from it was discharged to the municipal wastewater treatment plant. The amount of this sink was estimated by multiplying the amount of wastewater produced during the injections (approx. 45'000 m^3) with the average SNFC concentration in the wastewater measured during the injections (1.5 mg/L, n=2). According to this estimation the output was 60 kg, which is about 3 % of the totally used SNFC.

(3) About 5 % of the injected SNFC leached into the groundwater. This was calculated from the SNFC concentrations found in piezometer 2 as follows. Piezometer 2 is located about 55 m downstream from the end of the injection zone in the tunnel and at a distance of around 10 m from a groundwater flow boundary. In-between these two boundaries the whole groundwater diameter was influenced by the construction activities and the groundwater flowed directly towards the piezometer, allowing to assume plug flow conditions. The groundwater discharge rate in the area between the end of the injection zone and flow boundaring was calculated according to Darcy's law: Q = A*K*dh/dl, with A = cross-sectional area = 20 m * 10 m = 200 m^2, K = aquifer hydraulic conductivity = $2*10^{-3}$ m/s (estimation based on pump tests), dh/dl = hydraulic gradient = 5 ‰ (calculated from piezometer levels). Darcy's law is governed by the parameter K, which has a variability between $1*10^{-5}$ and $9*10^{-3}$ m/s in the Zurich aquifer.

For the mass flux calculation the groundwater discharge rate was multiplied with the measured N15S concentration in the piezometer and integrated over the breakthrough period of about 320 d. N15S was chosen because of its biological persistency. The concentrations of the biodegradable components (monomers) were estimated based on their relative amount to N15S in the technical SNFC mixture. Based on the results of the sorption experiments it was assumed that all monomers and the dimers were washed out to the same extent as N15S, the trimers and the tetramers only to an extent of 20 % and 10 %, respectively and that all other oligomers did not leach at all. These calculations resulted in a total of 140 kg of leached SNFC. This means that approx. 7 % of the SNFC was washed out. The calculations repeated with the dimer (persistent as well) instead of N15S yielded about 60 kg SNFC washed out (only about 3 %). This is probably due to the slightly different leaching characteristics of the two components. The average of the two calculations, 5 % of leached SNFC, was used.

(4) Some of the SNFC components are biodegradable. Based on earlier research it was assumed that about 20 % of the leached SNFC is persistent [9]. These are the components N15S, dimer, trimer and tetramer which in laboratory degradation experiments showed not to be biodegradable within 195 days. For the construction site presented here this means that about 100 kg of pure SNFC was leached into the aquifer and 20 kg of it remain in the aquifer (see Figure 4.5).

The most relevant result of the mass flux calculations is the leached SNFC amount (sink 3) since this is the main environmental concern. Hence, it was important to know if the calculated 5 % can be confirmed for other similar construction sites. Therefore, the results from the Zurich tunnel construction site were compared with estimations made at the Baregg tunnel construction site. Here, the SNFC from cement suspension injections leached into a groundwater of a very low flow velocity. The groundwater was sampled close to the construction site (at a distance of only 5-10 m) and SNFC concentrations of up to 230 µg/L were measured. The leached SNFC amount was calculated according to the volume approach of a box model based on the peak concentration in one piezometer. This rough estimation resulted in a total of 5 kg SNFC leached into the groundwater at the Baregg construction site. This corresponds to 1 % of leached SNFC. The comparison of the results of the two construction sites is difficult because of different groundwater flow

velocities and different SNFC concentrations in the cement suspension. Still, the results confirm that the amount of SNFC leached from fresh cement injections into aquifers is not negligible.
With these mass flux calculations it was possible to quantify the amount of SNFC which leached from fresh cement from two construction sites in direct contact with the groundwater. To some extent, this information can be extrapolated for the risk assessment of other polar organic construction chemicals with similar application. It can be inferred that they also leach to an amount around 5 %.

4.4 Conclusions

Mass flux calculations showed the transport behavior of one class of polar organic construction chemicals at two tunnel construction sites in contact with groundwater. Such calculations infer that polar organic construction chemicals leach into the environment in considerable amounts. This behavior emphasizes the need for an assessment of the environmental risk of toxic and persistent construction chemicals, before their application in aquifers. Additionally, by-products, impurities, and degradation products need to be considered in the risk assessment.
At the Zurich construction site, SNFC leached from fresh cement and concentrations of up to 58 μg/L were found in the aquifer at a distance of about 55 m from the construction site. With mass flux calculations it was shown that about 5 % of the injected SNFC leached into the aquifer. The leached components are the monomers and the oligomers up to the tetramers. Higher condensed oligomers (chainlength ≥ 5) appear to sorb strongly and are thus not washed out of the cement. Furthermore, laboratory sorption experiments indicated sorption onto the aquifer material being responsible for the different transport behavior of the dimer, trimer and tetramer in the groundwater. Due to the low SNFC concentrations in the groundwater and the low toxicity of SNFC no major impact on the Zurich aquifer is expected.

Acknowledgments

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5

Examples for SNFC Release into the Environment, Risk Assessment, and Practical Implications

5.1 Introduction

The source "application" was investigated in Chapter 3 and 4. The focus was on SNFC contamination of *groundwater* resulting from tunnel construction activities. In this chapter *wastewater* of a construction site is presented as another type of "application" source. Relevant amounts of SNFC can be expected to enter the aquatic environment through this pathway. The investigated construction site is a tunnel construction site where SNFC was used in shotcrete for the fixation of the walls of a shaft. The wastewater was discharged into a river after pretreatment.
Another focus of this chapter is on the source "production" which is also supposed to be significant. For this part investigations at a manufacturing plant synthesizing SNFC were made. The wastewater from reactor cleaning is discharged into an industrial wastewater treatment plant (WWTP) with biological treatment. The influent and effluent of the industrial WWTP was sampled and the amount of released SNFC was calculated. These investigations also revealed more detailed information on biodegradability of SNFC in adapted biological treatment systems.
Having determined the major sources for SNFC in the environment and the resulting environmental concentrations, the question should be asked whether the environment is impacted significantly by these concentrations. Are the measured environmental SNFC concentrations a risk for aquatic life or human health? The tools for answering this question are based on risk assessments methodology. Therefore, the compartment was chosen, in which the compounds of interest partition according to their physico-chemical properties. Then, the concentrations of the compound in the relevant environmental compartments have to be measured or estimated, leading to the exposure concentrations PEC (predicted environmental concentration). The PECs are compared with the predicted no effect concentration (PNEC) of the most sensitive organism in the relevant compartment. A PNEC is extrapolated from a lowest observable effect concentration (LOEL) that is determined under laboratory conditions and an assessment factor. Finally, the ratio between PEC and PNEC indicates the expected risk. Ratios below 1 lead to the conclusion that no risk for the environment is expected, while ratios above 1 indicate a potential risk. In this chapter a risk assessment for the water-soluble SNFC was performed

for the aquatic environment. The relevant compartments were groundwater and surface water. The risk for humans taking up drinking water from groundwater contaminated with SNFC and for aquatic organisms in surface water was assessed.
The studies in this chapter conclude the determination of the major SNFC sources and pathways. Since the sources and pathways are known specific measures against SNFC release into the environment can be taken. Possible measures are described in Chapter 5.4. These measures can partly be adapted for polar organic construction chemicals in general.

5.2 Experimental Section

Chemicals, instruments used and HPLC-FLD methods are described in Chapter 2. The following quality assurance parameters were obtained for construction site wastewater (more details in Appendix A3): Recoveries for SNFC compounds were, depending on the compounds, between 55 and 99 %, the relative standard deviations were between 4 and 12 %, the limits of detection between 0.01 and 0.13 μg/L and the limits of quantification between 0.02 and 0.44 μg/L. For the river water samples the results of the quality control experiments of groundwater were taken. For the samples of the WWTP no quality control experiments were made. Hence, only semi-quantitative conclusions were drawn from the results.

5.2.1 Tunnel Construction Site

The investigated location is a tunnel construction site of the Swiss Federal Railways. A new north-south-connection through the Gotthard is being built at the base of the mountain. The tunnel will have a length of about 57 km. To decrease the construction time, a shaft of 800 m depth was dug out in Sedrun (canton Graubünden), which allows to advance from the middle of the tunnel towards both its portals. The shaft in Sedrun was excavated by blasting and cement was placed on its walls. The bleeding water of the cement mixed with water from the drilling work and minor amounts of drained groundwater was pumped from the shaft bottom into a tank. From the tank, the water was released into the tunnel drain, where it was diluted with seeping groundwater and wastewater from the on-site cement paste mixing place. The collected waters were pretreated before discharge into the river Vorderrhein. The treatment consisted of neutralization, coagulation, and sedimentation in an artificial pond. The effluent of the pond entered the river Vorderrhein with a continuous flow of about 30 L/sec.

SNFC had been used until August 1999 for the fluidization of the cement, then it was substituted by another product based on sulfonated melamine formaldehyde condensates. Two weeks after the substitution of SNFC samples were collected from the tank at the top of the shaft, from the influent of the treatment plant, from the artificial pond and from the river Vorderrhein up- and downstream of the discharge point of the artificial pond. The samples were collected in amber glass bottles, immediately stabilized by adding 1 % v/v of formaldehyde solution and then stored at 4°C.

5.2.2 Manufacturing Plant and Industrial WWTP

The investigated manufacturing plant produces chemicals for the textile industry, including SNFC used by the textile industry. The wastewater from reactor cleaning is retained in a basin with a volume of 160 m^3. From the basin the wastewater flows continuously into an industrial WWTP (22.5 m^3/d) before being discharged into the sewerage system. The treatment plant consists of a bioreactor based on membrane technology. Samples of the treatment plant influent and effluent were taken during the week after the cleaning of the reactor previously used for SNFC production. The samples were stabilized on-site by adding 1 % v/v of formaldehyde solution and then stored at 4°C.

5.3 Results and Discussion

5.3.1 SNFC Release from Construction Site Wastewater

Table 5.1 shows the SNFC concentrations found at the five sampling points two weeks after the substitution of SNFC by another product. The concentrations are not representative for SNFC application, since SNFC was not applied during the two previous weeks. This is the reason why no mass flux calculations can be made. However, interesting semi-quantitative and qualitative conclusions can be inferred.

Table 5.1: *Total SNFC Concentrations (Sum Monomers and Oligomers)*

	[µg/L]
tank at the top of the shaft	30.1
influent treatment plant	3.2
artificial pond	2.4
river upstream of the effluent of the artificial pond	< LOD
river downstream of the effluent of the artificial pond	0.7

The relative concentrations of the different SNFC components in the samples provide some information on the behavior of SNFC in the wastewater system (see Chapter 3 and 4). The distribution pattern ("fingerprint") of the monomers in all contaminated samples of the wastewater system was similar to the one in the technical SNFC mixture, indicating that there was no biological activity in the wastewater system (see Chapter 3, the individual biodegradability of the SNFC components would yield a changed distribution pattern). Possibly biodegradation was prevented in the wastewater due to high pH and/or an excess of salts. However, the distribution pattern of the oligomers in the wastewater samples was different to the one in the technical SNFC mixture. As observed earlier (see Chapter 4) only oligomers with chainlength of up to four units were found. This relates to the strong adsorption ability of higher condensed oligomers which consequently do not leach from the cement.

This field study shows that SNFC can enter the aquatic environment by way of the source "application". The same SNFC components were found in the wastewater of this site as in the affected groundwater investigated earlier (Chapter 3 and 4). In the groundwater some of these components, mostly monomers, were degraded. The same should be true in the river water. However, no samples were taken from the river Vorderrhein at a longer distance from the discharge point to confirm this. Even so, the leached oligomers and the persistent naphthalene-1,5-disulfonate are expected to cause permanent pollution of the river as seen in the groundwater studies.

5.3.2 SNFC Release from Production Wastewater

The SNFC concentrations found in the influent of the industrial WWTP decreased from about 50 mg/L at the day of the reactor cleaning to about 20 mg/L after five days (Table 5.2). The concentration decrease is due to continuous output of SNFC into the treatment plant and no SNFC input into the basin (no SNFC was produced during the week investigated).

***Table 5.2**: Total SNFC Concentrations (Sum Monomers and Oligomers) in the Industrial WWTP*

days after reactor cleaning	SNFC Influent [mg/L]	SNFC Effluent [mg/L]
0	48.9	2.5
1	53.3	4.9
2	50.5	4.7
3	39.9	4.4
4	16.1	3.7
5	23.4	2.5

Closer analysis of the influent samples of the industrial WWTP revealed that the synthesized product is similar to the SNFC used as superplasticizers. The product consisted also of several mono- and disulfonated monomers and of oligomers of up to 17 units. Their relative concentrations reflected the distribution pattern known from superplasticizers. The distribution pattern of the SNFC components in the influent samples stayed constant over the whole week, giving evidence that no SNFC was biodegraded in the basin (see Chapter 3).

The SNFC concentrations in the samples of the effluent of the biological treatment plant decreased from 4.9 mg/L to 2.5 mg/L, indicating an SNFC elimination. N1S and N2S were never detected in the effluent (concentrations < 0.1 mg/L) and on the fourth day after the reactor cleaning N16S and N26S could also no longer be detected (Figure 5.1). After the fifth day the N17S and N27S concentrations decreased relative to the N15S concentrations. This sequence is typical for biodegradation and was already observed in laboratory experiments and in the field (Chapter 3). Oligomers, which were found to be persistent in the laboratory and in the field (Chapter 3), were only detected up to the tetramer. The elimination of the oligomers can not be inferred to be caused by biodegradation, despite of the adapted biological treatment system. The relative amounts of N15S and the oligomers in the effluent samples stayed constant over the whole week. Biological activity would change the pattern since the degradation kinetics is most probably compound specific. The elimination is rather expected to be based on sorption, since in adsorption experiments it was shown that oligomers with chainlengths of more than four units exhibit enhanced adsorption ability (Chapter 4). This is in accordance to similar investigations by Wolf et al. [1]. There, the removal of the higher condensed oligomers is attributed to destabilizing effects of polyvalent cations, inducing flocculation, coagulation, or sorption onto the activated sludge. The analysis of digest and fresh sludge confirmed the occurrence of SNFC in sludge [2].

The fact that SNFC components were found in the effluent of the SNFC production plant indicates that even adapted biological systems are not able to degrade N15S and SNFC oligomers. In literature, also flocculation with Al^{3+} salt was found to remove only oligomers with chainlength of more than seven units to more than 90 % and the monomers and dimers showed insignificant removal [2]. This infers that SNFC components can pass the WWTP unchanged and hence, reach the aquatic environment.

Effectively, sulfonated naphthalenes were detected in the effluent of WWTPs earlier [2, 3].

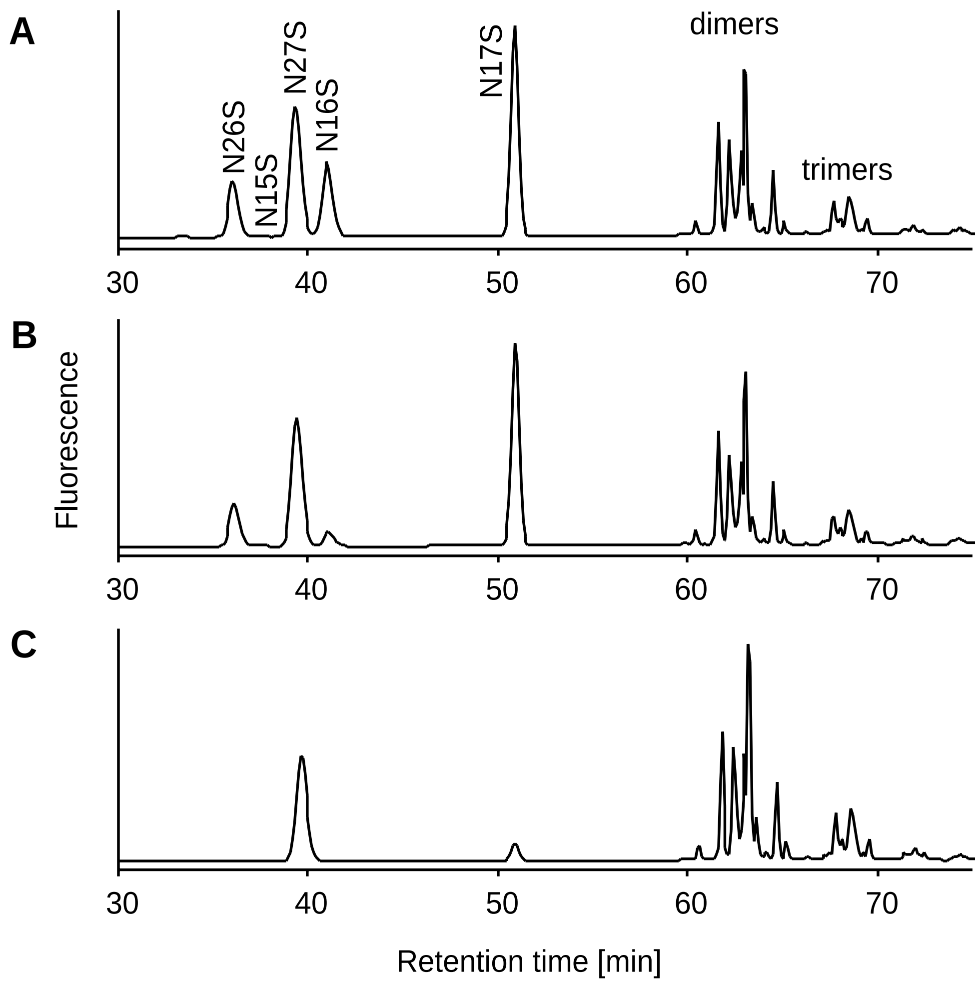

Figure 5.1: *HPLC-FLD chromatograms of the effluent of the biological treatment one day (A), two days (B) and five days (C) after reactor cleaning. The signal for N15S is not visible due to the scale.*

Also for the source "production" mass flux calculations were made in order to estimate the importance of this source. The calculations were based on the following assumptions: SNFC was released into the wastewater basin with the reactor cleaning water within hours. The released SNFC was diluted homogeneously in the basin, which at that time was full (160 m^3).

The effluent of the basin (22 m^3/d) was assumed to be not relevant for SNFC output during the hours of release and dilution of the cleaning water and the SNFC concentration in the wastewater sample after cleaning (50 mg/L) to be representative for the SNFC concentration in the basin and could then be multiplied by the basin volume for obtaining an estimation of the lost SNFC through the wastewater. This calculation resulted in about 8 kg of SNFC in the wastewater system, which accounted for 0.2 % of the production volume. When considering the elimination in the WWTP, the percentage of produced SNFC which reached the environment would even be lower. If biodegradation would be the only elimination process in the WWTP, 0.04 % of the production volume would enter the environment (see Chapter 3). A comparison of this percentage with the leaching at the source "application", where about 5 % reaches the environment (Chapter 4) showed that the 1 at the source "production" is about two orders of magnitudes lower. For absolute amounts (instead of percentages) detailed knowledge about the different applications of SNFC would have to be available. If more than 1 % of the produced SNFC is used for applications as described in Chapter 4, the source "application" would still be the prevailing source.

With the study of the industrial WWTP it could be shown that the persistent SNFC from the source "production" can be discharged into the environment. Through dilution in the sewerage system the SNFC concentrations in the environment resulting from the source "production" can be in the range of µg/L [2]. This implicates no direct environmental risk. Still, the source "production" is relevant due the absolute amount released into the environment and due to the persistency and mobility of the SNFC components. In contrast to the source "application" which occurs only during a limited time, this is a continuously operating SNFC source.

5.3.3 Risk Assessment of SNFC from Superplasticizers

Environmentally relevant parameters for SNFC are presented in Chapter 1.1.3. Shortly, SNFC are well soluble in water and thus have a low bioaccumulation potential. Their adsorption potential to soils and sediments increases with longer chainlength. Biodegradation is the main removal pathway for monomeric SNFC components in the environment. The toxicity of SNFC is low.

The environmental risk assessment was performed for SNFC in affected groundwater and surface water receiving (construction site) wastewater. These are the two compartments, in which the highest concentrations of the released SNFC are expected when their physico-chemical properties and the results obtained in this project are taken into account.

The environmental risk of SNFC in surface water was estimated based on the following worst case scenario for aquatic organisms:

The construction site wastewater with the highest measured SNFC concentrations (1.5 mg/L) is released directly into the surface water. The other investigated cases where construction site wastewater and production wastewater were released into the WWTP and subsequently into surface water were assumed to result in lower environmental SNFC concentrations due to biodegradation, flocculation and dilution in the WWTP. The PNEC (predicted no effect concentration) was estimated based on the lowest LC_{50} value for fish (100 mg/L) [4]. Use of the recommended assessment factor of 1000 [5] gave a PNEC of 100 µg/L for the toxicity for freshwater organisms. The PEC (predicted environmental concentration) was calculated from the construction site wastewater concentration of 1.5 mg/L divided by 10 for dilution with surface water. The dilution with 10 shall represent a very small river for this worst case scenario. The obtained PEC of 150 µg/L divided by the PNEC was above 1 (1.5). This means that for this worst case scenario aquatic life in the river could be impacted.

For the groundwater the risk for human health caused by uptake of SNFC with drinking water was assessed. For the calculation of the daily intake the realistic worst case was used, i.e. the highest measured SNFC concentrations in the groundwater (230 µg/L, field site Baregg) were taken and it was assumed that the groundwater is used as drinking water without further treatment. Assuming that an adult of 70 kg drinks 2 L of water (groundwater) per day, a daily intake of 7 µg/kg was obtained. The NEL (no effect level) for acute toxicity was estimated based on oral toxicity of SNFC for rats. The LD_{50} value of > 5'000 mg/kg body weight [6] was divided by the recommended assessment factor of 100 [7]. This resulted in a NEL of > 50 mg/kg body weight. The ratio of the daily intake and the

NEL (equivalent to the PEC-PNEC ratio) was far below 1 (> 10^{-4}). Thus, the construction activities resulted in groundwater concentrations which represent no acute risk to human health.

In conclusion, the released SNFC has no major impact on the environment because of low toxicity and low environmental concentrations although for a worst-case-scenario minor impact could be expected. Still, some SNFC components are persistent. In a complete environmental risk assessment, this fact would lead to high spatial ranges, which means that the persistent SNFC components are transported over wide distances through the aquatic environment [8]. Thus, it must be considered that the hazard potential is not just restricted to the nearest surroundings of the SNFC sources.

5.4 Practical Implications

In this study, the major sources for one representative of polar organic construction chemicals in the environment are identified and the major pathways for entering the environment are described. Hence, measures against the release of such chemicals into the natural waters can be taken depending on the individual situation at the construction site or production plant. Examples for such measures are:

- Production and application of the least toxic and least persistent product: If several products exist for the same application, the choice of the product should include environmental criteria. Toxic and persistent products should only be produced and applied were absolutely necessary. Their environmental behavior should then be carefully assessed and monitored.
- Closing water cycles: The wastewater of cement paste mixing places should be reused for the production of concrete or cement suspensions if the quality of the water fulfills the needs for cement suspensions.
- Avoiding the contact of fresh cement with groundwater: Because polar construction chemicals leach primarily from fresh cement, the contact with groundwater during the setting phase should be avoided. This could be attained by a temporary lowering of the groundwater table. However, this measure causes problems such as ground settlements and influences the use of the groundwater.

- Funnel-and-gate-system: The contaminated groundwater downstream of the construction site should be separated from the surrounding groundwater with impermeable walls vertical to the flow direction and should be directed through adsorptive material such as carbon black in order to remove the contaminants.
- Sanitation wells for intervention measures: If prevention measures such as suggested above can not be taken, intervention measures combined with monitoring should be considered. An example for intervention measures are sanitation wells downstream of the construction site which allow to pump contaminated groundwater.

The first and second measures are the most important and should be followed whenever possible. All other measures could avoid contamination but influence the groundwater physically. It is not easy to decide which influence on the environment is minor. It depends on the single situation and on the toxicity of the construction chemicals. Hence, the decision has to be made for the individual situation.

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6

Conclusions

This study aimed at contributing to the environmental risk assessment of polar organic construction chemicals. The focus of the study was on the aquatic environment since these compounds are well soluble in water. Sources for SNFC in the environment were elucidated and the behavior of SNFC in natural waters was evaluated. The following conclusions can be drawn:

- Synchronous scan fluorimetry is a suitable tool for the routine analysis of SNFC in aqueous samples, allowing the monitoring of SNFC in the aquatic environment.
- SNFC from wastewater of manufacturing plants and construction sites can be found in ambient waters. Persistent SNFC components (N15S and oligomers) reach the aquatic environment through wastewater treatment plants. If discharged directly as in the case of construction activities in the aquifers, also biodegradable SNFC components are detected in natural waters.
- When fresh cement containing SNFC is applied at construction sites in the aquifer, SNFC leach into the groundwater. At one investigated construction site, SNFC leached to an extent of about 5 % of the total applied amount. The maximum concentration found in the groundwater was 230 µg/L at a distance of 5-10 m from the construction site.
- Only monomeric SNFC and oligomers with chainlengths of up to four units leach into the groundwater. The oligomers with longer chainlengths strongly adsorb onto cement and can not be found in the groundwater.
- In the groundwater, most monomeric SNFC is degraded within 195 days. N15S and the leached oligomers are persistent. The persistent fraction of the leached SNFC is about 20 %.

It can be concluded that the sources "application" and "production" are relevant for the occurrence of SNFC in the aquatic environment. Since the environmental concentrations and the toxicity of SNFC are low, risk assessments indicated no major impact on the environment and on human health. Still, some SNFC components are persistent and contaminate surface water, groundwater and eventually drinking water since they are not easy to remove from raw water.

This study provides an example for the environmental impact assessment for polar organic construction chemicals. The findings emphasize the need for assessing the environmental risk of construction chemicals when applied in aquifers in order to prevent incidents with more toxic representatives (such as [1-3]). In addition, byproducts, impurities and degradation products should be considered in the assessment, since their behavior can be different from the principal chemical. In the white paper of the European Communities on the strategy for a future chemicals policy, the persistent representatives of organic chemicals are classified within the chemicals with hazard properties giving rise to very high concern [4].

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A

Appendix

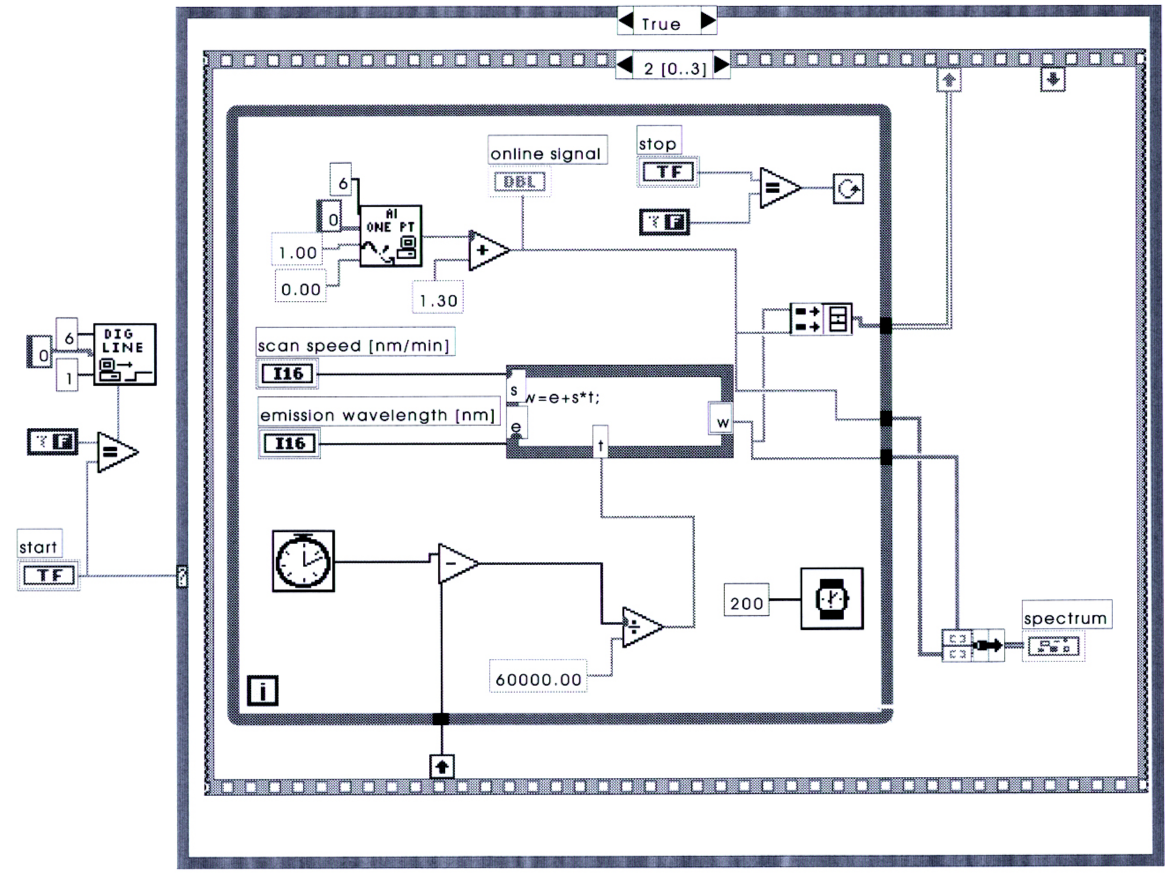
A1	Data Acquisition with LabVIEW (Chapter 2)
A2-A3	Quality Assurance Parameters of the HPLC-FLD Method (Chapter 2)
A4	Situation at the Baregg Field Site (Chapter 3)
A5-A7	SNFC Concentrations in the Groundwater of the Baregg Field Site (Chapter 3)
A8	Characterization of the Different Groundwaters (Chapters 3 and 4)
A9-A10	Situation at the Zurich-Thalwil Field Site (Chapter 4)
A11-A21	SNFC Concentrations in the Groundwater of the Zurich-Thalwil Field Site (Chapter 4)
A22	Situation at the Sedrun Field Site (Chapter 5)

A1: *Data Acquisition with LabVIEW*

The output signal of the fluorescence spectrometer (0 - 1 V) was acquired on a Macintosh Centris 650 with an additional NI-MIO-16 card (AC/DC converter) from National Instruments. With this card it was possible to trigger the start of the acquisition and the scanning of the fluorescence spectrometer. The further conversion of the signal was programmed with LabVIEW 5.0.1.

Acquisition mode:
Input parameters are scan speed and emission wavelength. They are used to convert the scanning time to the emission wavelength. Five data pairs per second (emission wavelength vs. signal voltage) are acquired. The data is displayed real-time. At the end of the measurement the data is written into a spreadsheet together with additional sample information (sample name, scaling factor, scanning mode, excitation wavelength and the date and time of the acquisition).

Analysis mode:
Data work up can be done offline. It allows manual integration of the signal.



Scheme A1: *Example for visualized programming with LabVIEW.*

Table A2: *Quality Control Parameters of the HPLC-FLD Method for Groundwater*

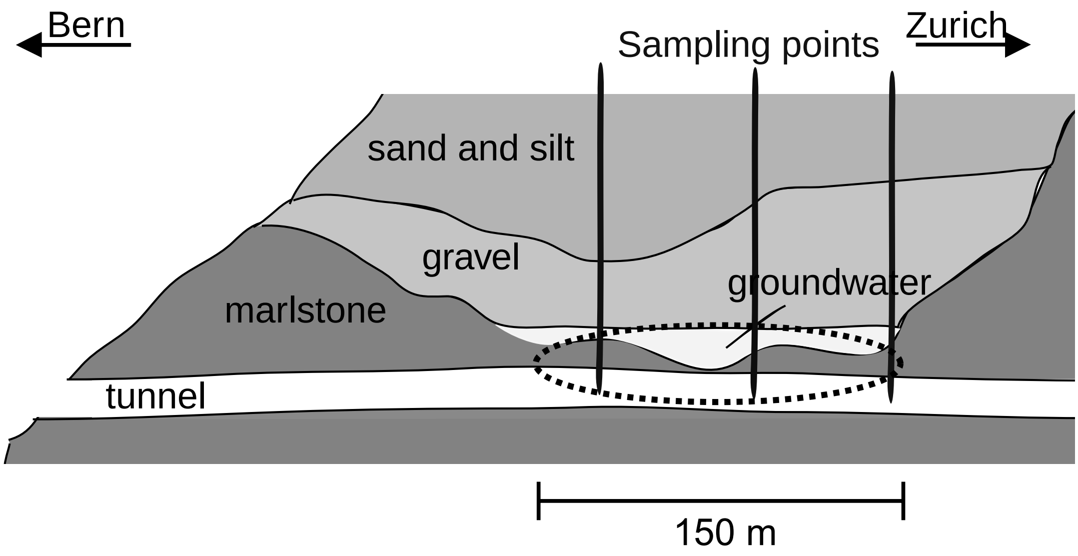
	recovery [%]	precision* [%]	LOD [µg/L]	LOQ [µg/L]
naphthalene-1-sulfonate	104	16	0.04	0.15
naphthalene-2-sulfonate	94	11	0.04	0.15
naphthalene-2,6-disulfonate	95	13	0.02	0.06
naphthalene-1,5-disulfonate	83	43	0.04	0.14
naphthalene-2,7-disulfonate	88	13	0.02	0.05
naphthalene-1,6-disulfonate	86	7	0.03	0.10
naphthalene-1,7-disulfonate	85	11	0.01	0.03
dimer	85	16	0.02	0.06
trimer	82	13	0.02	0.08
tetramer	75	12	0.03	0.11
pentamer	70	14	0.02	0.08
hexamer	64	20	0.02	0.06
heptamer	59	26	0.01	0.05

* = relative standard deviation * $t_{n-1,\,95\,\%}$ (n=8)

Table A3: *Quality Control Parameters of the HPLC-FLD Method for Construction Site Wastewater*

	recovery [%]	precision* [%]	LOD [µg/L]	LOQ [µg/L]
naphthalene-1-sulfonate	99	18	0.10	0.34
naphthalene-2-sulfonate	95	15	0.10	0.36
naphthalene-2,6-disulfonate	n.a.	n.a.	n.a.	n.a.
naphthalene-1,5-disulfonate	n.a.	28	0.13	0.44
naphthalene-2,7-disulfonate	95	16	0.03	0.10
naphthalene-1,6-disulfonate	86	17	0.07	0.23
naphthalene-1,7-disulfonate	88	10	0.01	0.02
dimer	79	12	0.01	0.03
trimer	68	11	0.08	0.26
tetramer	69	11	0.05	0.17
pentamer	65	11	0.03	0.11
hexamer	60	11	0.03	0.11
heptamer	55	13	0.03	0.09

n.a. = not analyzed, * = relative standard deviation * $t_{n-1,\,95\,\%}$ (n=8)



Scheme A4: *Situation at the Baregg field site. From a pilot tunnel a cement suspension was injected into the marked area (-----) over a length of about 150 m.*

Table A5: *SNFC Concentrations [µg/L] at Piezometer KB 2 of the Baregg Field Site*

Date	N1S	N2S	N26S	N15S	N27S	N16S	N17S	dimer	trimer	total SNFC
08/15/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
09/19/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
09/26/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
10/03/00	<0.15	0.2	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	0.2
10/11/00	19.4	65.6	6.4	9.1	24.0	68.8	28.6	10.6	0.3	232.8
10/17/00	0.7	2.2	0.2	0.4	0.8	2.5	0.9	0.3	<0.08	8.0
10/24/00	0.2	0.6	0.1	<0.14	0.2	0.8	0.3	0.1	<0.08	2.3
10/31/00	<0.15	0.2	<0.06	<0.14	<0.05	0.2	0.1	<0.06	<0.08	0.5
11/06/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	0.04	<0.06	<0.08	0.04
11/13/00	<0.15	0.2	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	0.2
11/20/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
11/27/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
12/04/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
12/12/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	0.04	<0.06		0.04
01/11/01	<0.15	0.4	<0.06	<0.14	<0.05	0.1	0.1	<0.06	<0.08	0.6
01/23/01	<0.15	0.4	<0.06	<0.14	<0.05	<0.10	0.04	<0.06	<0.08	0.44
02/06/01	<0.15	0.3	<0.06	<0.14	<0.05	<0.10	0.03	<0.06	<0.08	0.33
02/19/01	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ

LOQ = Limit of quantification

Table A6: *SNFC Concentrations [µg/L] at Piezometer KB 3 of the Baregg Field Site*

Date	N1S	N2S	N26S	N15S	N27S	N16S	N17S	dimer	trimer	total SNFC
08/15/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
09/19/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
09/26/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
10/03/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
10/11/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
10/17/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
10/31/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
11/06/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
11/13/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
11/20/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
11/27/00	<0.15	0.3	<0.06	<0.14	0.1	0.5	0.2	0.1	0.1	1.3
12/04/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
12/12/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
12/18/01	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
01/11/01	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
01/23/01	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
02/06/01	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
02/19/01	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ

LOQ = Limit of quantification

Table A7: *SNFC Concentrations [µg/L] at Piezometer KB 4 of the Baregg Field Site*

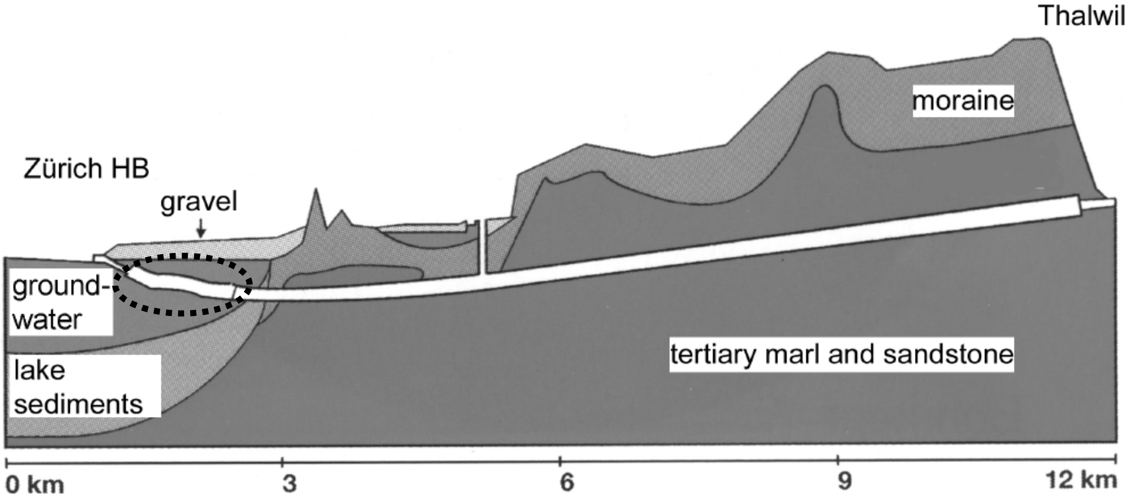
Date	N1S	N2S	N26S	N15S	N27S	N16S	N17S	dimer	trimer	total SNFC
08/15/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
09/19/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
09/26/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
10/03/00	<0.15	0.27	<0.06	<0.14	<0.05	<0.10	0.04	<0.06	<0.08	0.31
10/11/00	14.1	56.6	5.0	6.6	17.5	52.3	25.8	15.9	2.4	196.0
10/17/00	3.1	10.1	0.9	1.7	3.3	10.4	4.9	2.7	0.4	37.5
10/24/00	14.9	53.6	4.7	6.7	15.9	48.8	23.1	12.8	1.8	182.3
10/31/00	16.0	53.6	4.4	6.9	15.7	48.6	23.0	11.0	2.0	181.2
11/06/00	10.1	0.8	3.8	5.5	14.7	45.9	21.2	9.3	1.4	112.7
11/13/00	1.9	0.3	3.4	4.8	13.3	41.4	19.6	9.1	1.2	95.0
11/20/00	<0.15	0.2	2.4	4.7	14.4	39.7	20.1	9.2	1.3	92.0
11/27/00	<0.15	<0.15	<0.06	4.3	14.2	1.7	19.6	9.2	1.2	50.2
12/04/00	<0.15	0.3	<0.06	4.4	12.5	0.3	18.4	8.2	1.2	45.3
12/12/00	<0.15	0.2	<0.06	4.2	11.8	0.3	17.5	7.2	1.1	42.3
12/18/00	<0.15	0.2	<0.06	4.1	11.7	0.4	17.2	7.0	1.1	41.7
01/11/01	<0.15	0.4	<0.06	3.6	5.5	0.4	15.3	6.6	1.0	32.8
01/23/01	<0.15	0.3	<0.06	3.5	0.1	0.5	15.1	6.5	1.0	27.0
02/06/01	<0.15	0.4	<0.06	3.4	0.2	0.4	15.3	6.7	0.8	27.2
02/19/01	<0.15	0.2	<0.06	3.2	0.3	0.3	13.9	6.0	0.7	24.6
03/21/01	<0.15	0.2	<0.06	2.7	0.3	0.2	11.3	5.0	0.7	20.4
04/10/01	<0.15	<0.15	<0.06	2.3	0.2	0.2	8.9	4.0	0.6	16.2
05/16/01	<0.15	<0.15	<0.06	1.9	<0.05	0.1	0.7	2.5	0.4	5.6
06/13/01	<0.15	<0.15	<0.06	1.7	<0.05	<0.10	0.6	2.2	0.4	4.9
08/21/01	<0.15	<0.15	<0.06	1.4	<0.05	0.1	0.3	1.8	0.4	4.00

LOQ = Limit of quantification

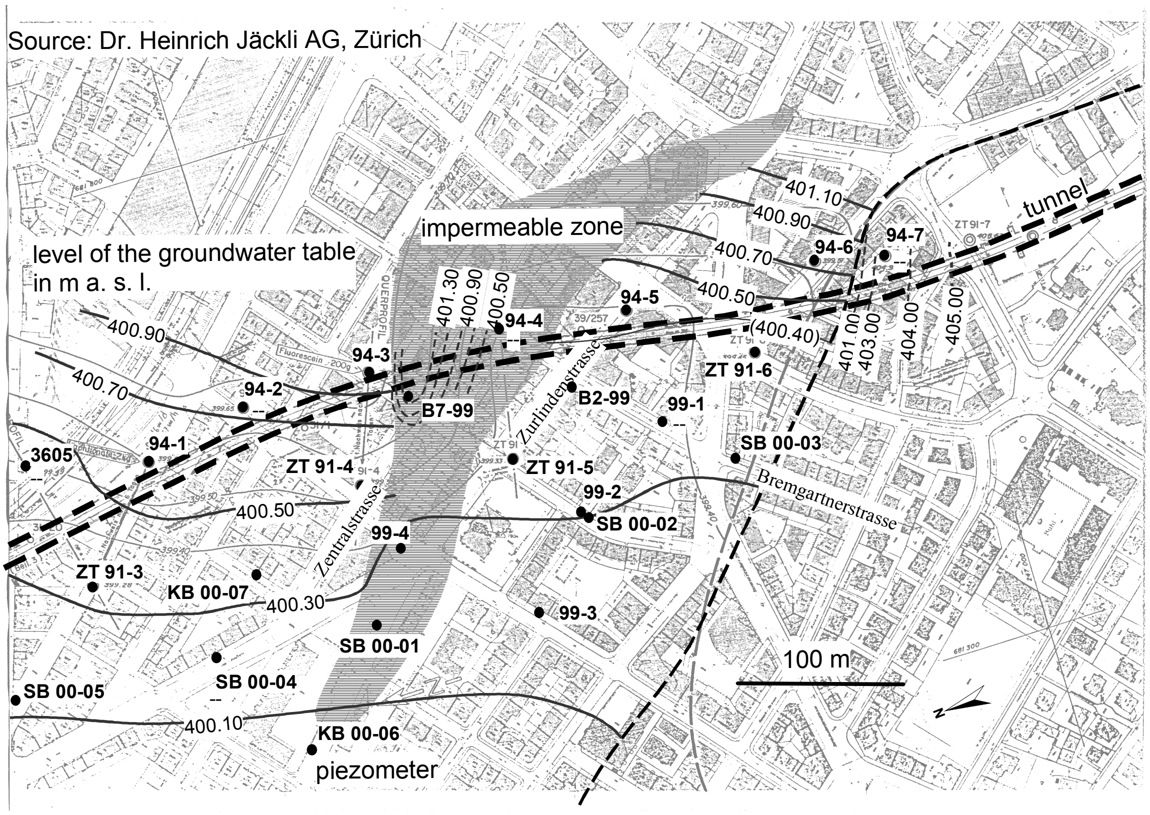
Table A8: *Characterization of the Different Groundwaters*

Parameter	Groundwater Baregg	Groundwater Zurich-Thalwil	Groundwater EAWAG
pH	7.53	7.35	7.4
Conductivity [µS/cm]	402	536	670
Chloride [mg/L]	3.9	18	18.4
Nitrite [mg/L]	<0.005	<0.005	<0.1
Nitrate [mg/L]	13	25	9.3
Sulfate [mg/L]	13	38	36.2
Ammonium [mg/L]	<0.01	<0.01	<0.1
DOC	0.26	0.45	1.8 (TOC)

DOC = dissolved organic carbon, TOC = total organic carbon



Scheme A9: *Situation at the Zurich-Thalwil field site. From a pilot tunnel a cement suspension was injected into the marked area (-----) over a length of about 400 m. Source: SBB.*



Scheme A10: *Groundwater situation at the Zurich-Thalwil field site. Source: Dr. Heinrich Jäckli AG.*

Table A11: *SNFC Concentrations [µg/L] at Piezometer KB ZT 91-3 of the Zurich-Thalwil Field Site*

Date	N1S	N2S	N26S	N15S	N27S	N16S	N17S	dimer	trimer	total SNFC
08/24/99	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
09/20/99	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	0.1	0.1	<0.08	0.2
02/21/01	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	0.1	<0.08	0.1

LOQ = Limit of quantification

Table A12: *SNFC Concentrations [µg/L] at Piezometer KB ZT 91-4 of the Zurich-Thalwil Field Site*

Date	N1S	N2S	N26S	N15S	N27S	N16S	N17S	dimer	trimer	total SNFC
08/24/99	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
09/20/99	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
12/17/99	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
05/09/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
06/06/00	<0.15	<0.15	<0.06	<0.14	0.1	0.3	0.1	<0.06	<0.08	0.5
07/04/00	<0.15	<0.15	<0.06	<0.14	0.1	<0.10	0.2	<0.06	<0.08	0.2
08/02/00	<0.15	<0.15	<0.06	4.0	<0.05	<0.10	1.3	0.9	<0.08	6.2
08/30/00	<0.15	<0.15	<0.06	5.5	<0.05	<0.10	1.9	1.4	<0.08	8.8
09/26/00	<0.15	<0.15	<0.06	11.0	0.1	<0.10	3.4	4.0	0.2	18.7
10/24/00	<0.15	<0.15	<0.06	14.1	<0.05	0.2	3.9	4.4	0.2	22.8
11/21/00	0.2	<0.15	<0.06	20.4	<0.05	0.4	5.5	8.7	0.6	35.8
02/21/01	<0.15	<0.15	<0.06	35.4	<0.05	0.6	0.1	17.7	3.5	57.7
03/27/01	<0.15	<0.15	<0.06	7.0	<0.05	0.1	<0.03	4.1	1.7	13.3
04/09/01	<0.15	<0.15	<0.06	7.4	<0.05	0.2	0.03	4.8	2.0	14.9
05/22/01	<0.15	<0.15	<0.06	6.3	<0.05	0.1	<0.03	2.5	1.0	10.2
06/07/01	<0.15	<0.15	<0.06	16.8	<0.05	0.3	<0.03	8.1	2.2	27.9
08/21/01	<0.15	<0.15	<0.06	8.5	<0.05	0.2	<0.03	3.7	1.5	14.3

LOQ = Limit of quantification

From 21.02.01 to 21.08.01 tetramer was found in concentrations of 0.4, 0.4, 0.5, 0.3, 0.5 and 0.4 µg/L.

Table A13: *SNFC Concentrations [µg/L] at Piezometer KB ZT 91-6 of the Zurich-Thalwil Field Site*

Date	N1S	N2S	N26S	N15S	N27S	N16S	N17S	dimer	trimer	total SNFC
08/24/99	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
09/20/99	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
11/22/99	<0.15	<0.15	<0.06	<0.14	0.1	0.3	0.1	<0.06	<0.08	0.5
12/17/99	<0.15	<0.15	<0.06	0.3	0.1	<0.10	0.2	0.1	<0.08	0.7
01/18/00	<0.15	<0.15	<0.06	1.7	<0.05	<0.10	0.4	0.3	<0.08	2.4
02/16/00	<0.15	<0.15	<0.06	0.4	<0.05	<0.10	0.1	0.2	<0.08	0.7
03/10/00	<0.15	<0.15	<0.06	0.4	<0.05	<0.10	0.1	0.2	<0.08	0.7
04/11/00	<0.15	<0.15	<0.06	1.4	<0.05	<0.10	0.4	0.3	<0.08	2.1
05/09/00	<0.15	<0.15	<0.06	0.5	<0.05	<0.10	0.1	0.1	<0.08	0.7
06/06/00	<0.15	<0.15	<0.06	0.6	<0.05	<0.10	0.1	0.1	<0.08	0.8
07/04/00	<0.15	<0.15	<0.06	1.4	<0.05	<0.10	0.2	0.2	<0.08	1.8
08/02/00	<0.15	<0.15	<0.06	1.8	<0.05	<0.10	0.2	0.3	<0.08	2.3
08/30/00	<0.15	<0.15	<0.06	2.8	<0.05	<0.10	0.2	0.6	<0.08	3.6
09/26/00	<0.15	<0.15	<0.06	3.2	<0.05	<0.10	0.1	1.1	0.1	4.5
10/24/00	<0.15	<0.15	<0.06	4.0	0.2	0.4	0.4	1.9	0.4	7.3
02/21/01	<0.15	<0.15	<0.06	0.4	<0.05	<0.10	<0.03	0.2	<0.08	0.6
05/22/01	<0.15	<0.15	<0.06	0.2	<0.05	<0.10	<0.03	0.1	<0.08	0.3

LOQ = Limit of quantification

Table A14: *SNFC Concentrations [μg/L] at Piezometer KB 99-1 of the Zurich-Thalwil Field Site*

Date	N1S	N2S	N26S	N15S	N27S	N16S	N17S	dimer	trimer	total SNFC
08/24/99	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
09/20/99	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
11/22/99	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
12/17/99	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
01/18/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
02/16/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
03/10/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	0.1	<0.06	<0.08	0.1
04/11/00	<0.15	<0.15	<0.06	0.9	<0.05	<0.10	0.3	0.3	<0.08	1.5
05/09/00	<0.15	<0.15	<0.06	1.1	<0.05	<0.10	0.3	0.3	<0.08	1.7
06/06/00	<0.15	<0.15	<0.06	1.3	<0.05	<0.10	0.4	0.4	<0.08	2.1
07/04/00	<0.15	<0.15	<0.06	1.3	<0.05	<0.10	0.3	0.5	<0.08	2.1
08/02/00	<0.15	<0.15	<0.06	0.9	<0.05	<0.10	0.2	0.3	<0.08	1.4
08/30/00	<0.15	<0.15	<0.06	0.9	<0.05	<0.10	0.1	0.3	<0.08	1.3
09/26/00	<0.15	<0.15	<0.06	1.0	<0.05	<0.10	0.1	0.3	<0.08	1.4
10/24/00	<0.15	<0.15	<0.06	1.2	<0.05	<0.10	0.1	0.4	<0.08	1.7
02/21/01	<0.15	<0.15	<0.06	0.8	<0.05	<0.10	<0.03	0.2	<0.08	1.0
05/22/01	<0.15	<0.15	<0.06	0.4	<0.05	<0.10	<0.03	0.1	<0.08	0.5

LOQ = Limit of quantification

Table A15: *SNFC Concentrations [μg/L] at Piezometer KB 99-2 of the Zurich-Thalwil Field Site*

Date	N1S	N2S	N26S	N15S	N27S	N16S	N17S	dimer	trimer	total SNFC
08/24/99	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
09/20/99	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
11/22/99	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
01/18/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
02/16/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
03/10/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
04/11/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
05/09/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
06/06/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
07/04/00	<0.15	<0.15	<0.06	0.4	<0.05	<0.10	0.1	<0.06	<0.08	0.5
08/30/00	<0.15	<0.15	<0.06	1.7	<0.05	<0.10	0.5	0.1	<0.08	2.3
09/26/00	<0.15	<0.15	<0.06	1.4	<0.05	<0.10	0.3	0.2	<0.08	1.9
02/21/01	<0.15	<0.15	<0.06	1.5	<0.05	<0.10	0.1	0.3	<0.08	1.9
05/22/01	<0.15	<0.15	<0.06	1.9	<0.05	<0.10	0.04	0.5	<0.08	2.4

LOQ = Limit of quantification

Table A16: *SNFC Concentrations [μg/L] at Piezometer KB B2-99 of the Zurich-Thalwil Field Site*

Date	N1S	N2S	N26S	N15S	N27S	N16S	N17S	dimer	trimer	total SNFC
08/24/99	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
09/20/99	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
11/22/99	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
12/17/99	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
01/18/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
02/16/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
03/10/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
04/11/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
05/09/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
06/06/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
07/04/00	<0.15	<0.15	<0.06	0.2	<0.05	<0.10	0.1	<0.06	<0.08	0.3
08/02/00	<0.15	<0.15	<0.06	0.6	<0.05	<0.10	0.2	0.2	<0.08	1.0
08/30/00	<0.15	<0.15	<0.06	0.2	<0.05	<0.10	0.1	<0.06	<0.08	0.3
09/26/00	<0.15	<0.15	<0.06	0.2	<0.05	<0.10	0.1	<0.06	<0.08	0.3
10/24/00	<0.15	<0.15	<0.06	0.2	<0.05	<0.10	0.1	<0.06	<0.08	0.3
02/21/01	<0.15	<0.15	<0.06	0.2	<0.05	<0.10	0.04	0.1	<0.08	0.3
05/22/01	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
08/21/01	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ

LOQ = Limit of quantification

Table A17: *SNFC Concentrations [μg/L] at Piezometer SB 00-02 of the Zurich-Thalwil Field Site*

Date	N1S	N2S	N26S	N15S	N27S	N16S	N17S	dimer	trimer	total SNFC
03/28/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
08/02/00	<0.15	<0.15	<0.06	1.1	<0.05	<0.10	0.3	0.1	<0.08	1.5
10/24/00	<0.15	<0.15	<0.06	1.5	<0.05	<0.10	0.2	0.2	<0.08	1.9
08/21/01	<0.15	<0.15	<0.06	1.2	<0.05	<0.10	<0.03	0.3	<0.08	1.5

LOQ = Limit of quantification

Table A18: *SNFC Concentrations [µg/L] at Piezometer SB 00-03 of the Zurich-Thalwil Field Site*

Date	N1S	N2S	N26S	N15S	N27S	N16S	N17S	dimer	trimer	total SNFC
03/28/00	<0.15	<0.15	<0.06	2.5	<0.05	<0.10	0.8	0.9	<0.08	4.2
04/11/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
05/09/00	<0.15	<0.15	<0.06	2.8	<0.05	<0.10	0.9	1.0	<0.08	4.7
06/06/00	<0.15	<0.15	<0.06	2.6	<0.05	<0.10	0.8	0.9	<0.08	4.3
07/04/00	<0.15	<0.15	<0.06	3.1	<0.05	<0.10	0.9	1.2	0.1	5.3
08/02/00	<0.15	<0.15	<0.06	4.5	<0.05	<0.10	1.0	2.2	0.2	7.9
08/30/00	<0.15	<0.15	<0.06	4.9	<0.05	<0.10	0.9	2.3	0.3	8.4
09/26/00	<0.15	<0.15	<0.06	5.0	<0.05	<0.10	0.8	2.3	0.3	8.4
10/24/00	<0.15	<0.15	<0.06	5.4	<0.05	<0.10	0.4	2.5	0.3	8.6
02/21/01	<0.15	<0.15	<0.06	4.0	<0.05	<0.10	<0.03	1.5	0.2	5.7
05/22/01	<0.15	<0.15	<0.06	2.3	<0.05	<0.10	<0.03	0.9	0.1	3.3
08/21/01	<0.15	<0.15	<0.06	1.9	<0.05	<0.10	<0.03	0.7	0.1	2.7

LOQ = Limit of quantification

Table A19: *SNFC Concentrations [µg/L] at Piezometer SB 00-04 of the Zurich-Thalwil Field Site*

Date	N1S	N2S	N26S	N15S	N27S	N16S	N17S	dimer	trimer	total SNFC
03/28/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
05/09/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
06/06/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
07/04/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
08/02/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	<0.06	<0.08	<LOQ
08/30/00	<0.15	<0.15	<0.06	0.6	<0.05	<0.10	0.2	0.1	<0.08	0.9
09/26/00	<0.15	<0.15	<0.06	0.9	<0.05	<0.10	0.2	0.1	<0.08	1.2
10/24/00	<0.15	<0.15	<0.06	1.2	<0.05	<0.10	0.3	0.3	<0.08	1.8
11/21/00	<0.15	<0.15	<0.06	1.2	<0.05	<0.10	0.3	0.3	<0.08	1.8
02/21/01	<0.15	<0.15	<0.06	0.5	<0.05	<0.10	<0.03	0.2	<0.08	0.7
05/22/01	<0.15	<0.15	<0.06	0.9	<0.05	<0.10	<0.03	0.2	<0.08	1.1
06/07/01	<0.15	<0.15	<0.06	0.9	<0.05	<0.10	<0.03	0.2	<0.08	1.1
08/21/01	<0.15	<0.15	<0.06	0.3	<0.05	<0.10	<0.03	0.1	<0.08	0.4

LOQ = Limit of quantification

Table A20: *SNFC Concentrations [µg/L] at Piezometer SB 00-05 of the Zurich-Thalwil Field Site*

Date	N1S	N2S	N26S	N15S	N27S	N16S	N17S	dimer	trimer	total SNFC
03/28/00	<0.15	<0.15	<0.06	<0.14	<0.05	<0.10	<0.03	0.1	<0.08	0.1

LOQ = Limit of quantification

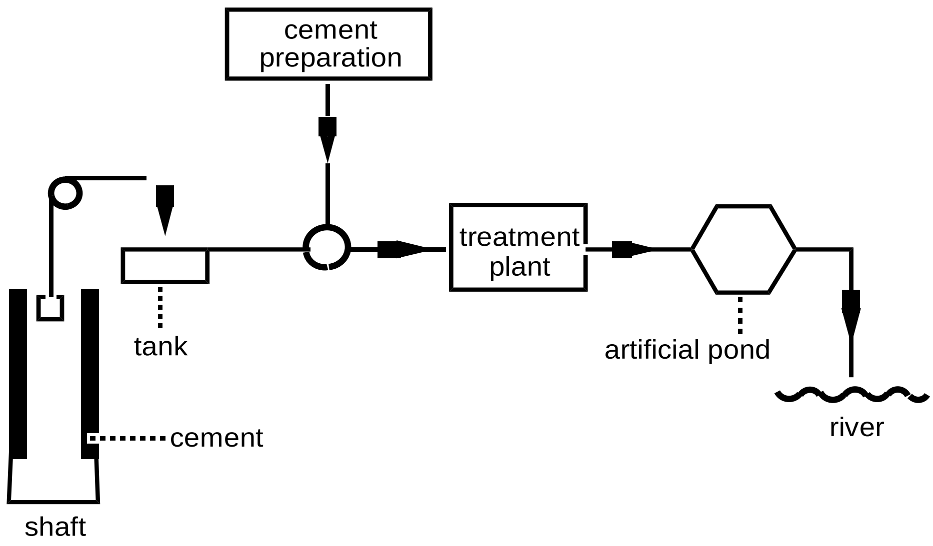
Table A21: *SNFC Concentrations [µg/L] at Piezometer SB 00-07 of the Zurich-Thalwil Field Site*

Date	N1S	N2S	N26S	N15S	N27S	N16S	N17S	dimer	trimer	total SNFC
02/21/01	<0.15	<0.15	<0.06	3.5	<0.05	<0.10	<0.03	1.7	0.6	5.9
05/22/01	<0.15	<0.15	<0.06	1.9	<0.05	<0.10	<0.03	0.6	0.1	2.6

LOQ = Limit of quantification

On the 21.02.01 tetramer was found in concentrations of 0.1 µg/L.

The groundwater of the piezometers 94-1, 94-2, 94-3, 94-4, 94-5, 94-6, B7-99, and 99-3 was only analyzed before the beginning of the construction activities (08/24/99 and 09/20/99) and the groundwater of the piezometers SB 00-01 and SB 00-06 was analyzed once (03/28/01). All SNFC concentrations were below the limit of quantification. At piezometer ZT 91-5 no SNFC was found over the whole observation period (08/24/99 - 05/22/01).



Scheme A22: *Situation at the Sedrun field site.*

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